



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade



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School of Computing

Department of Computer Science and Engineering

Digital Forensics

Integrated Course Theory

Lab Record (213CSE4307)

Student Name: -

Register Number: -

Section & slot: -



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BONAFIDE CERTIFICATE

Bonafide record of work done by _____
of _____ in _____
_____ during even/odd semester in academic year
_____.

Staff In-charge

Submitted to the practical Examination held at Kalasalingam University,
Krishnankoil on _____

REGISTER NUMBER

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Internal Examiner

External Examiner

School of Computing
Department of Computer Science and Engineering
Academic Year: 2026-2027

Subject Name: Digital Forensics

Year / Sem: III Year – V Semester

Course Handling Faculty: Dr. K. Venkatesh

Section: 24S16

S.NO	DATE	TITLE OF THE PROGRAM	MARKS	SIGN
1		Create a forensic image of a storage device using Guymager or DC3DD, and verify its integrity using MD5 and SHA-256 hash values.		
2		Recover deleted or damaged files from a storage device using TestDisk and Foremost.		
3		Capture and analyze network traffic to identify suspicious activities using Wireshark.		
4		Analyze email headers and identify phishing or email spoofing attacks using Mail Header Analyzer (MHA) or MX Toolbox Email Header Analyzer.		
5		Create a digital forensic case, import evidence, recover deleted files, and generate investigation reports using Autopsy.		
6		Examine file systems, metadata, timelines, and deleted files using The Sleuth Kit (TSK).		
7		Acquire and analyze volatile memory from a Linux or Android environment using LiME, and investigate memory artifacts using Volatility 3.		
8		Detect hidden information in image files using zsteg and Stegseek (or Steghide for embedding/extraction).		
9		Monitor and analyze running processes, system calls, and file activities using Sysdig or Strace to identify suspicious system behavior.		
10		Perform basic malware analysis using REMnux by examining file hashes, strings, YARA rules, and Indicators of Compromise (IOCs).		

Ex. No. 1	Evidence Acquisition Using Access Data FTK Imager
Date	

Description

Forensic Toolkit or FTK is a computer forensics software product made by Access Data. This is a Windows based commercial product. For forensic investigations, the same development team has created a free version of the commercial product with fewer functionalities. This FTK Imager tool is capable of both acquiring and analyzing computer forensic evidence.

The evidence FTK Imager can acquire can be split into two main parts. They are:

- Acquiring volatile memory
- Acquiring non-volatile memory (Hard disk)

There are two possible ways this tool can be used in forensics image acquisitions:

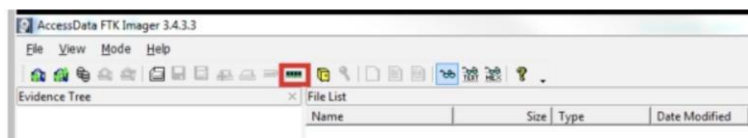
- Using FTK Imager portable version in a USB pen drive or HDD and opening it directly from the evidence machine. This option is most frequently used in live data acquisition where the evidence PC/laptop is switched on.
- Installing FTK Imager on the investigator's laptop.

In this case the source disk should be mounted into the investigator's laptop via write blocker. The write blocker prevents data being modified in the evidence source disk while providing read-only access to the investigator's laptop. This helps to maintain the integrity of the source disk.

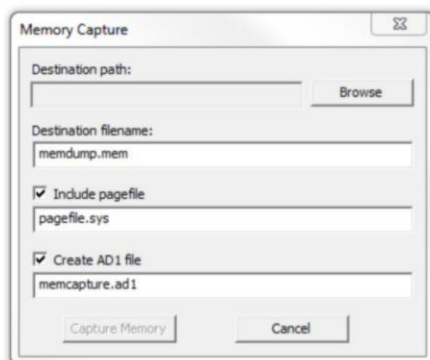
Acquiring volatile memory using FTK Imager

The FTK Imager tool helps investigators to collect the complete volatile memory (RAM) of a computer. The following steps will show you how to do this.

Open FTK Imager and navigate to the volatile memory icon (capture memory).



Navigate to the destination location where you need to save the captured volatile memory and create a file name.

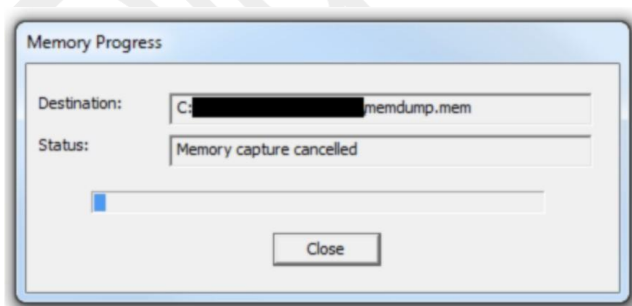


NOTE: This tool provides options to include pagefile and AD1 files when acquiring the volatile memory.

Pagefile: The pagefile (pagefile.sys) is used in Windows operating systems as volatile memory due to limitation of physical random-access memory (RAM). It is located under the “C” partition ready to use as volatile memory when the existing RAM capacity is exceeded. So this file can have quite a bit of valuable data when considering the volatile memory. Therefore, it is recommended to capture and collect this file in the acquisition.

AD1 file: AD1 is the FTK imager image file. The investigator has the option to create an AD1 file for later use.

Clicking the “capture memory” button will start acquiring the volatile memory.

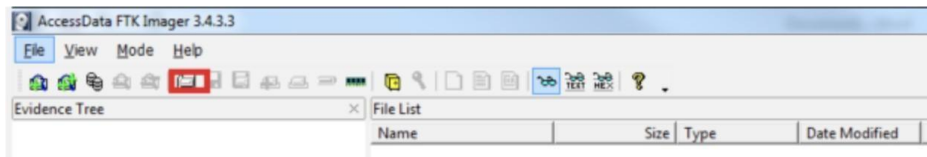


NOTE: Once the acquisition has completed, the destination folder will have the acquired memory with the file extension of “.mem”.

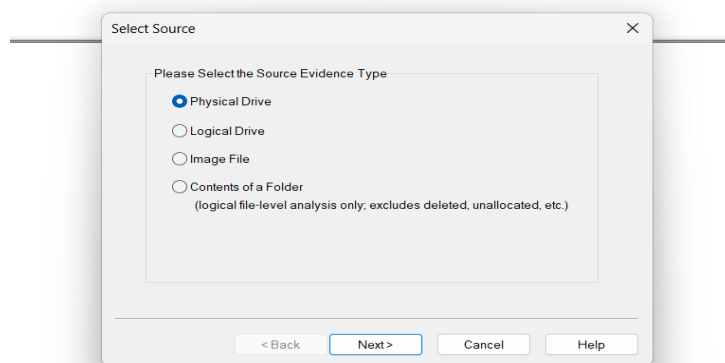
Acquiring non-volatile memory (Disk Image) using FTK Imager

As previously stated, this same tool can be used to collect a disk image as well.

Open FTK Imager and navigate to “Create Disk Image”.



Now select the source that you need to acquire.



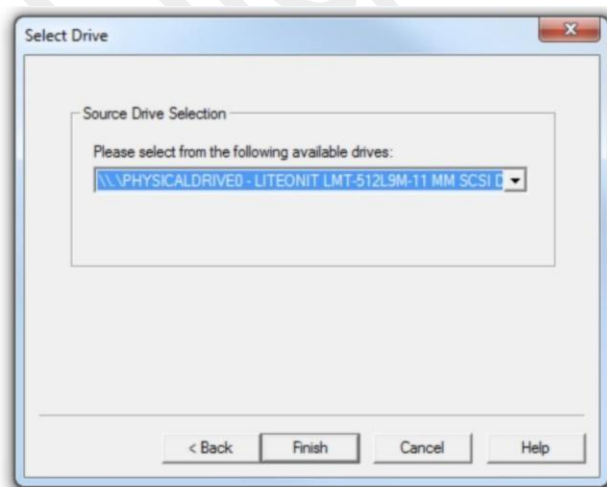
NOTE: FTK Imager is capable of acquiring physical drives (physical hard drives), logical drives (partitions), image files, contents of a folder, or CDs/DVDs. Investigators can connect external HDDs into the collection computer

via write blocker and use the “logical drive” option to select the mounted HDD as a partition.

Collecting Physical Drives

Select the “Physical Drive” option.

Select the drive you need to acquire and click “Finish”.



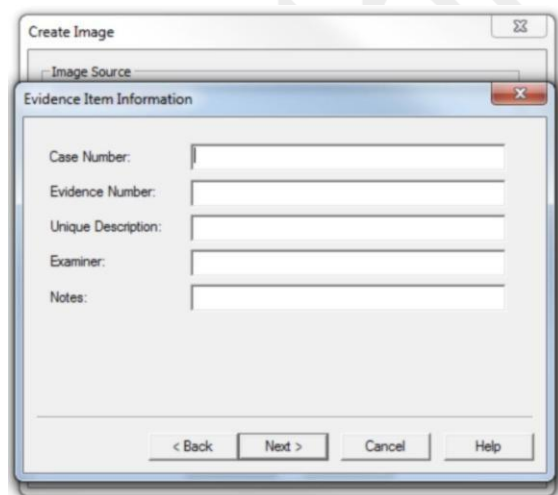
Raw (dd): This is the image format most commonly used by modern analysis tools. These raw file formatted images do not contain headers, metadata, or magic values. The raw format typically includes padding for any memory ranges that were intentionally skipped (i.e., device memory) or that could not be read by the acquisition tool, which helps maintain spatial integrity (relative offsets among data).

SMART: This file format is designed for Linux file systems. This format keeps the disk images as pure bitstreams with optional compression. The file consists of a standard 13-byte header followed by a series of sections. Each section includes its type string, a 64-bit offset to the next section, its 64-bit size, padding, and a CRC, in addition to actual data or comments, if applicable.

E01: this format is a proprietary format developed by Guidance Software's EnCase. This format compresses the image file. An image with this format starts with case information in the header and footer, which contains an MD5 hash of the entire bit stream. This case information contains the date and time of acquisition, examiner's name, special notes and an optional password.

AFF: Advance Forensic Format (AFF) was developed by Simson Garfinkel and Basis Technology. Its latest implementation is AFF4. The goal is to create a disk image format that does not lock the user into a proprietary format that may prevent them from being able to properly analyze it.

Now enter the case details.



The screenshot shows a 'Create Image' dialog box with the 'Evidence Item Information' tab selected. The tab contains the following fields:

- Case Number: [Text Input Field]
- Evidence Number: [Text Input Field]
- Unique Description: [Text Input Field]
- Examiner: [Text Input Field]
- Notes: [Text Input Field]

At the bottom of the dialog box, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'.

Add an image destination (where the image file will be saved), image file name and fragment size.

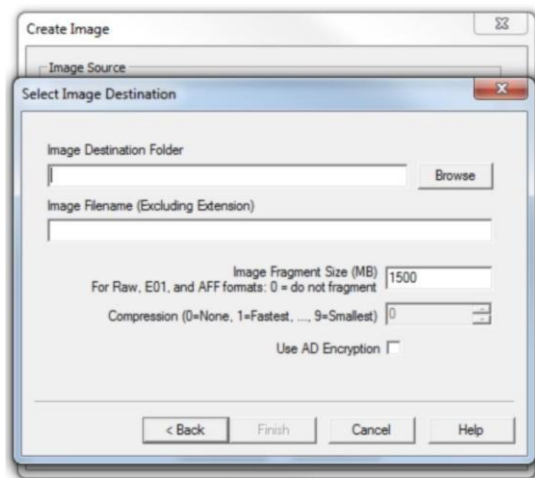
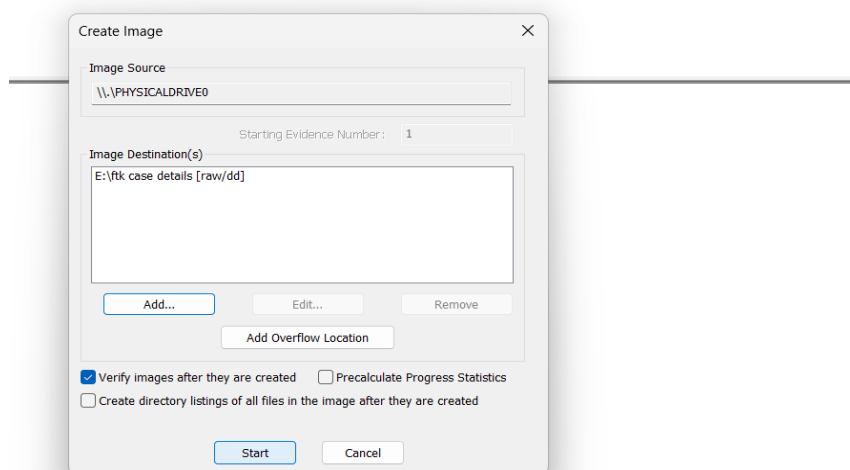
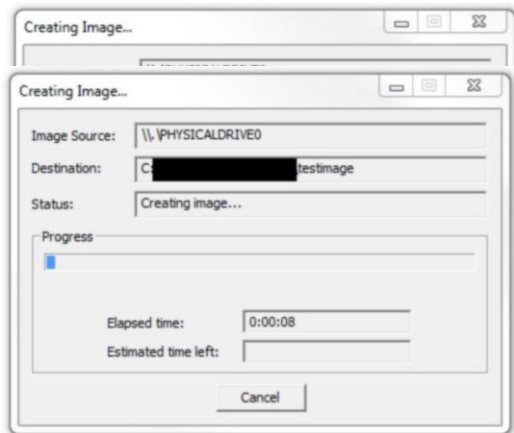


Image Fragment Size (MB): this option will separate the image file into multiple images and save them in the same destination. If you need only one file instead of creating multiple fragmented images, you must set the image fragment size to “0”.

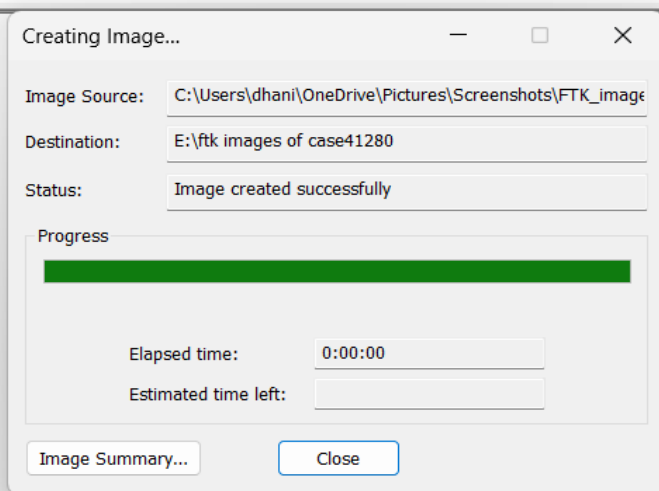
Select the “verify images after they are created” option. This will verify the hash values once the image has created. In order to ensure integrity, it is recommended to use this option. However, this will increase the time taken to acquire your evidence, especially if you’re dealing with a large disk image size.

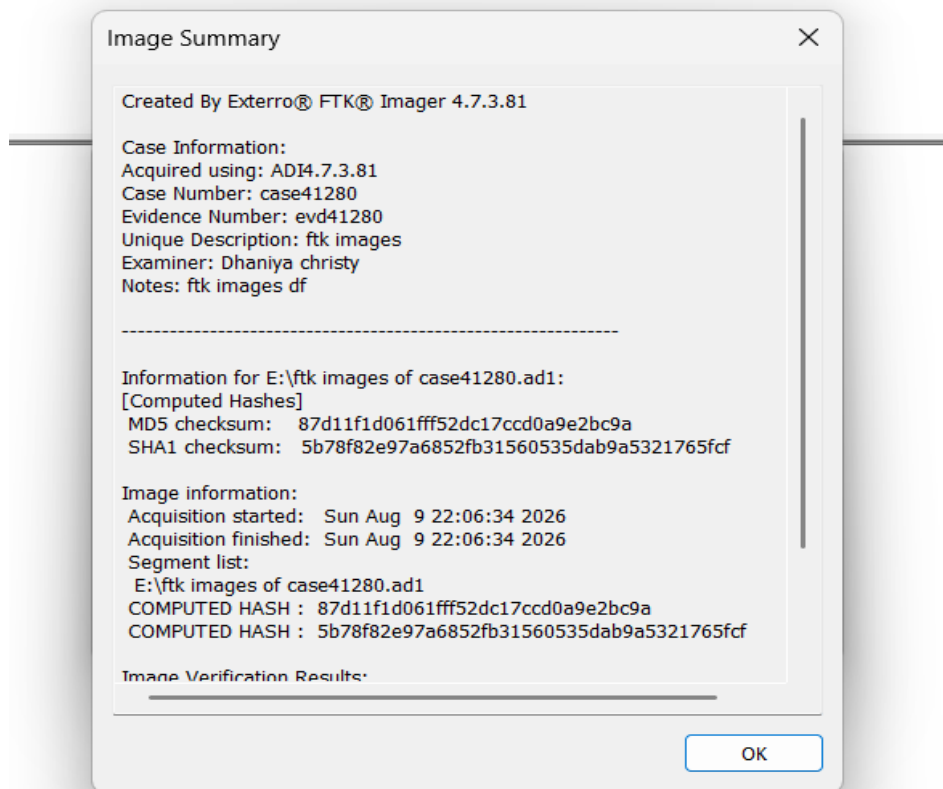




Click “start” to start acquiring.

Once acquiring is complete, it will create a text file including all the information it has acquired.





Hash values are matched.

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

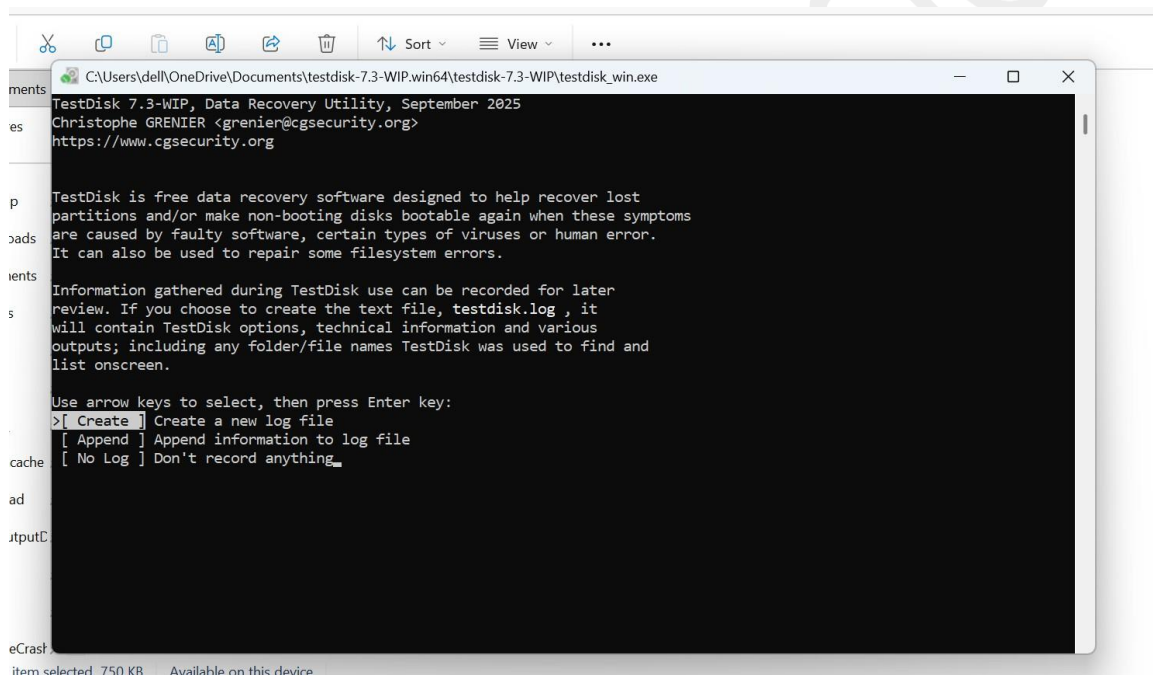
Evidence was successfully acquired using FTK Imager, including volatile memory and disk imaging. The acquired image was verified using hash values, confirming the integrity of the forensic evidence.

Ex.No.2	Recover deleted or damaged files from a storage device using Test Disk
Date	

Description:

TestDisk step by step to recover a missing partition and corrupted one. Log

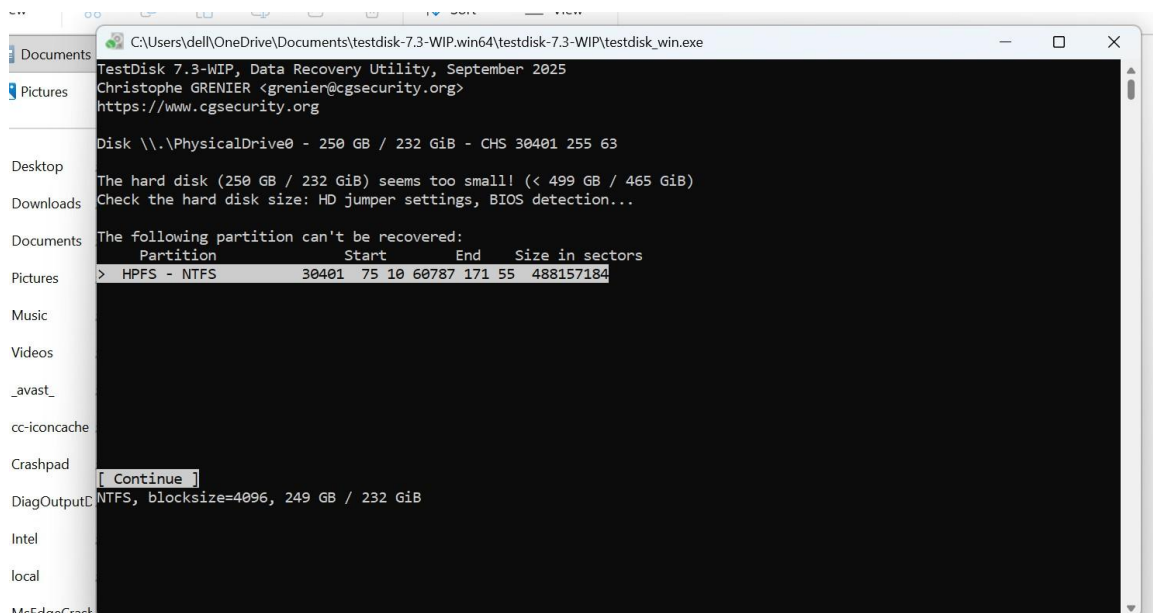
Creation



- Choose Create to instruct Testdisk to create a **log file** containing technical information and messages, unless you have a reason to append data to the **log** or you execute TestDisk from read only media and must create the **log** elsewhere.
- Choose None if you do not want messages and details of the process to be written into a **log file** (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk Selection

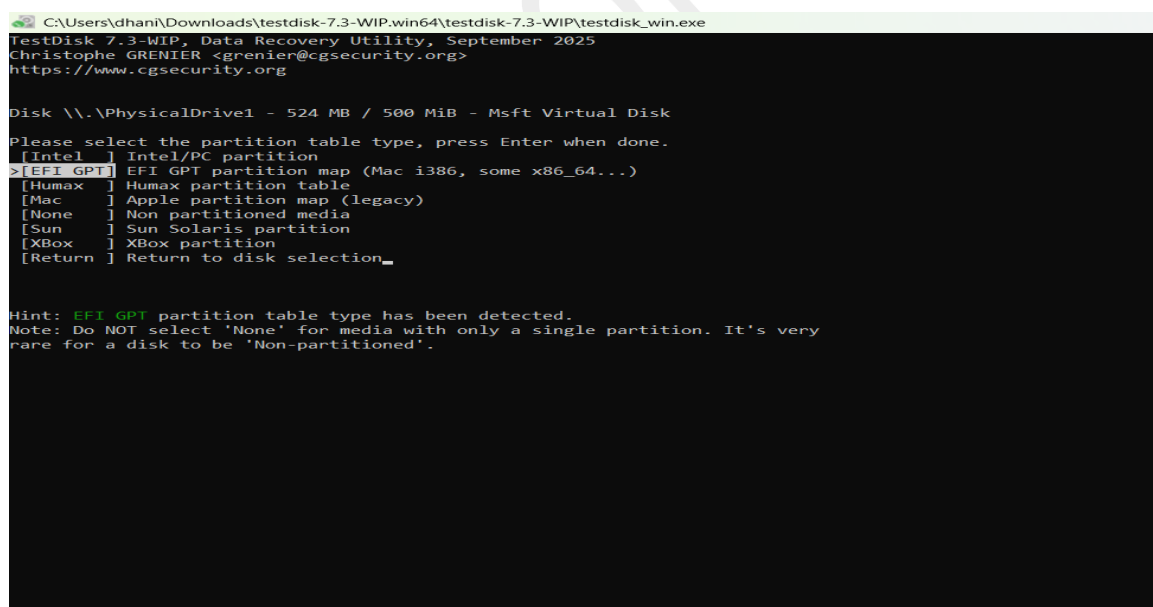
All hard drives should be detected and listed with the correct size by TestDisk:



- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.
- If available, use raw device /dev/rdisk* instead of /dev/disk* for faster data transfer.

Partition Table Type Selection

TestDisk displays the partition table types.



- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current Partition Table Status

```

C:\Users\dnani\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
ristophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 524 MB / 500 MiB - Msft Virtual Disk
CHS 63 255 63 - sector size=512

Analyse ] Analyse current partition structure and search for lost partitions
Advanced ] Filesystem Utils
Geometry ] Change disk geometry
Options ] Modify options
Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.

```

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

```

C:\Users\del\OneDrive\Documents\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive0 - 250 GB / 232 GiB - CHS 30401 255 63
Current partition structure:
Partition      Start      End      Size in sectors
1 P EFI GPT    0 0 2 267349 89 4 4294967295
Bad sector count:
No partition is bootable

+=Primary bootable P=Primary L=Logical E=Extended D=Deleted
[Quick Search] [Backup] Try to locate partition

```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

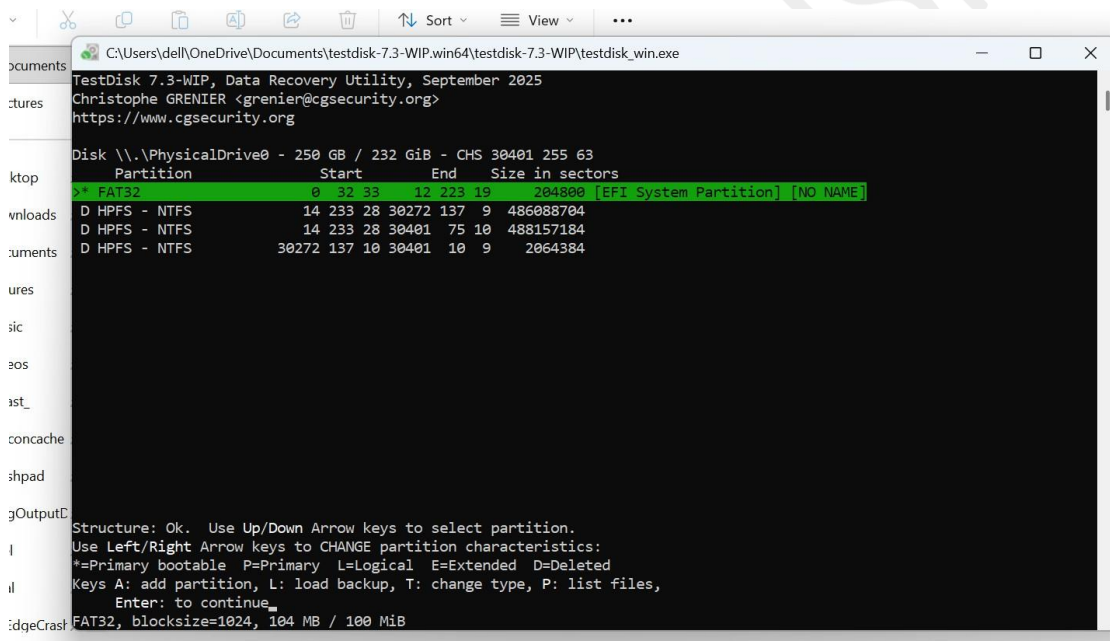
Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

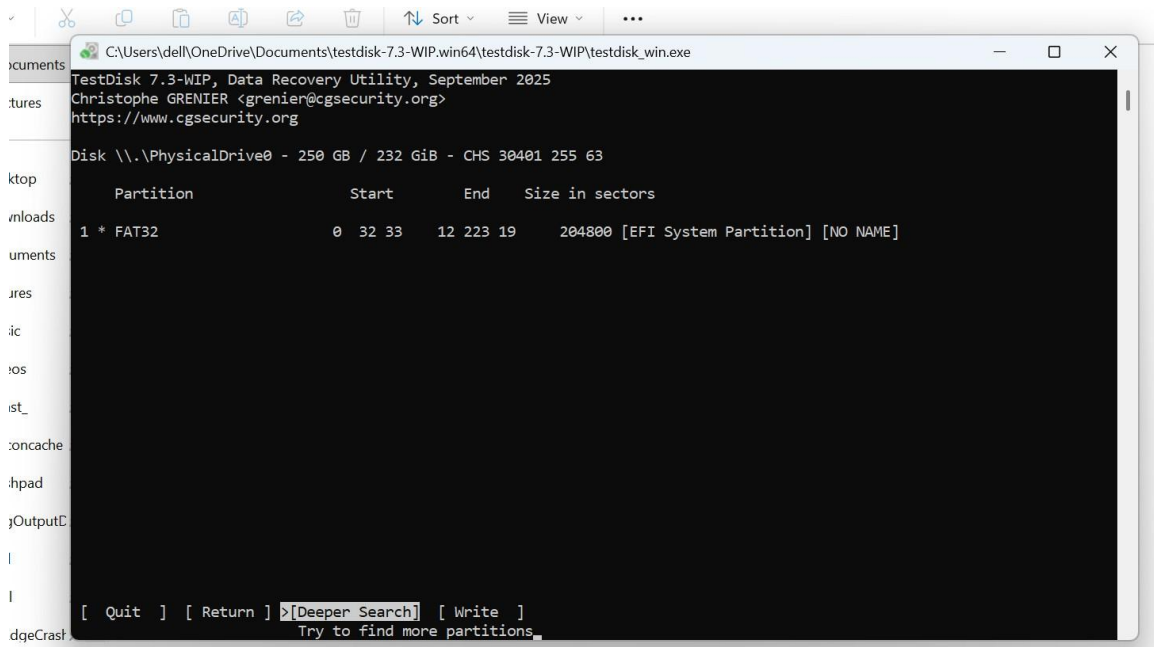
Quick Search for partitions

TestDisk displays the first results in real time.



- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).
- All directories and data are correctly listed.
- Press Enter to proceed.

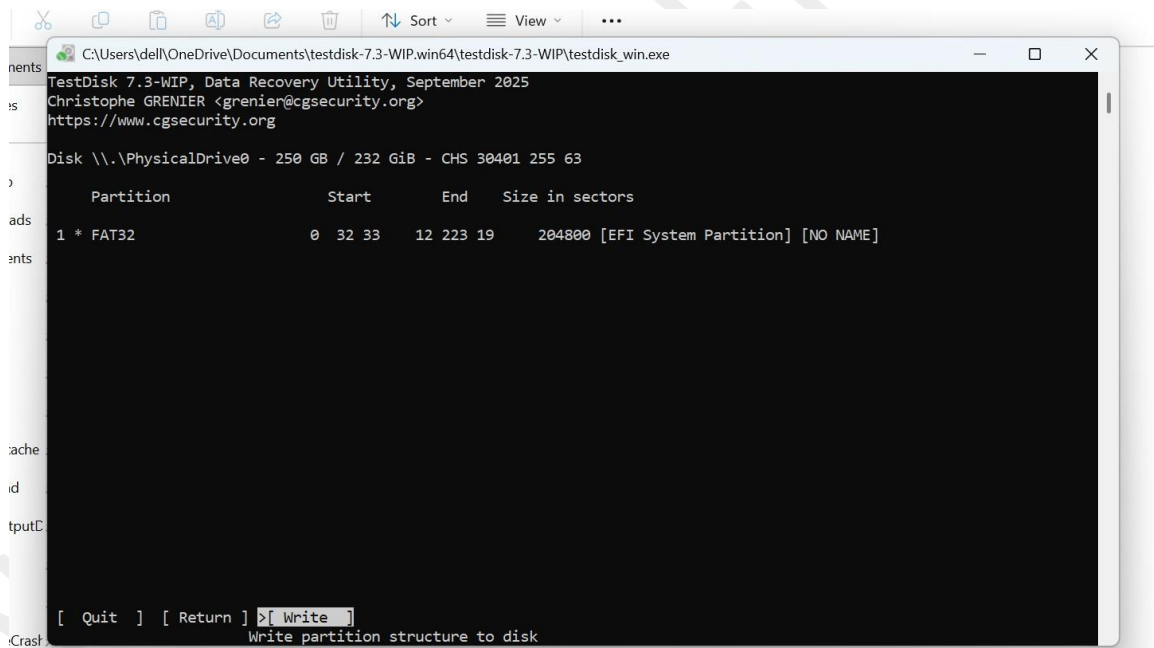
Save the partition table or search for more Partitions?



Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

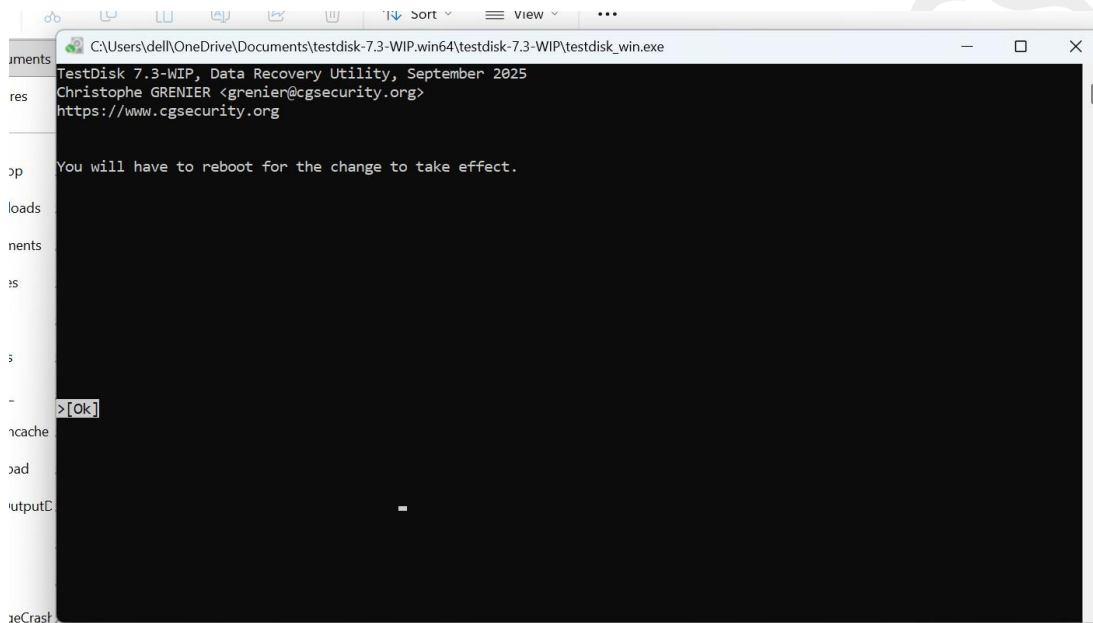


- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

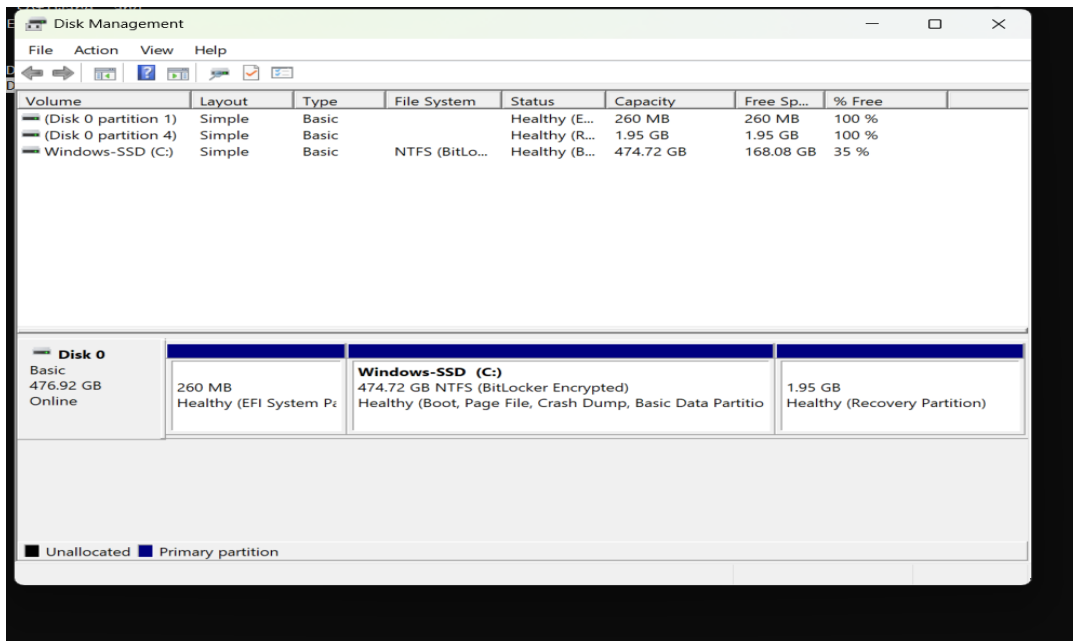
Now, the partitions are registered in the partition table.

NTFS Boot sector Recovery

The boot sector of the first partition named Partition 1 is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.



- TestDisk displays ***You have to restart your Computer to access your data*** so press Enter a last time and reboot your computer.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

TestDisk successfully identified and recovered the missing partition and repaired the damaged NTFS boot sector.

The recovered partition structure was restored, allowing access to the stored data after restarting the computer.

Ex.No.3	Wireshark
Date	

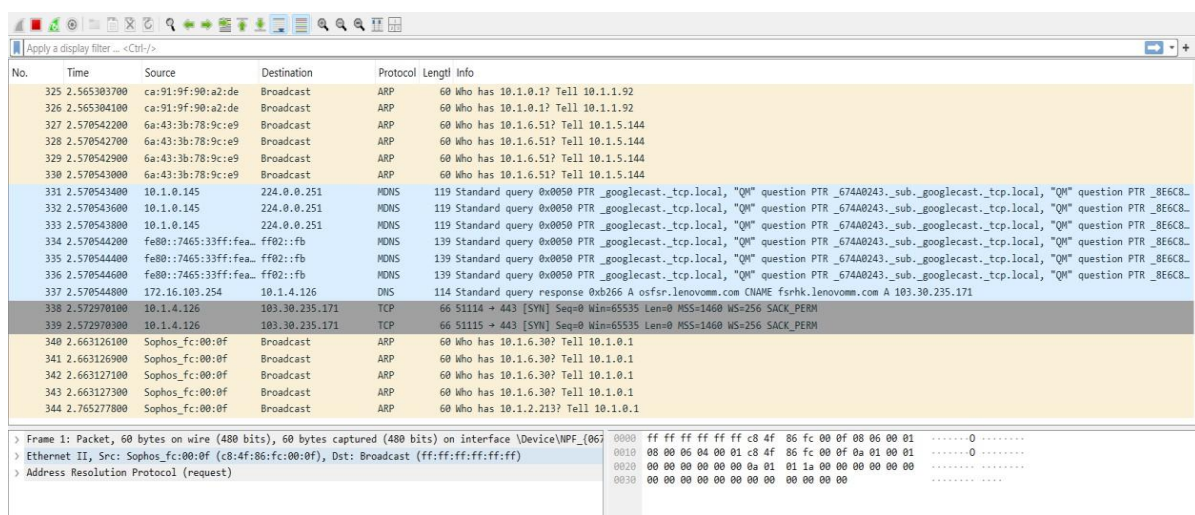
Password Capturing

Wireshark can capture not only passwords but any type of information transmitted over the network: usernames, email addresses, personal information, etc. As long as we can capture network traffic, Wireshark can sniff passing passwords.

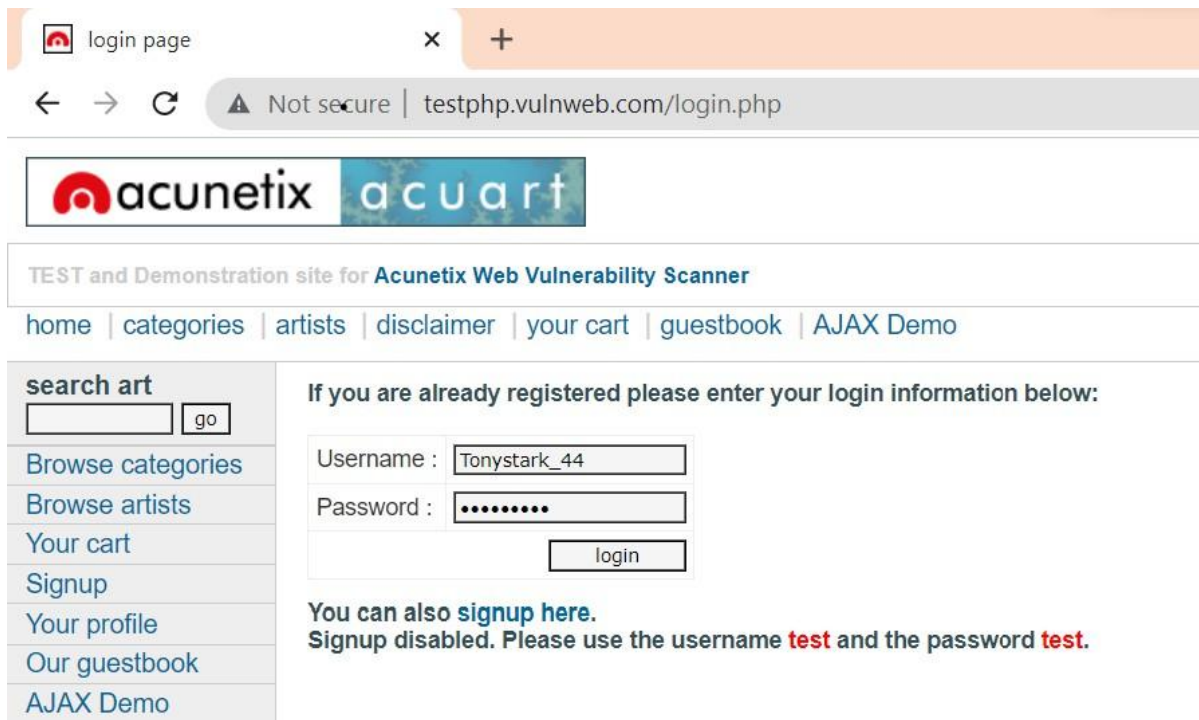
In sniffing can include passwords for various protocols such as HTTP, FTP, Telnet, etc. the captured data can be used to troubleshoot network problems, but can also be used maliciously to gain unauthorized access to sensitive information.

So, here we will see how we can capture the password using the Wireshark network capture analyzer. and see the outputs of the following steps.

Step 1: First of all, open your Wireshark tool in your window or in Linux virtual machine. and start capturing the network. suppose I am capturing my wireless fidelity.



Step 2: After starting the packet capturing we will go to the website and login the credential on that website as you can see in the image.



Step 3: Now after completing the login credential we will go and capture the password in Wireshark. for that we have to use some filter that helps to find the login credential through the packet capturing.

Step 4: Wireshark has captured some packets but we specifically looking for HTTP packets. so in the display filter bar we use some command to find all the captured HTTP packets. as you can see in the below image the green bar where we apply the filter.
http

No.	Destination	Protocol	Length	Info
1	127.0.0.1	HTTP	756	GET / HTTP/1.1
2	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)
3	127.0.0.1	HTTP	723	GET / HTTP/1.1
4	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)
5	127.0.0.1	HTTP	649	GET /favicon.ico HTTP/1.1
6	127.0.0.1	HTTP	649	GET /favicon.ico HTTP/1.1
7	127.0.0.1	HTTP	936	POST /login HTTP/1.1 (application/x-www-form-urlencoded)
8	127.0.0.1	HTTP	44	HTTP/1.0 200 OK (text/html)
9	127.0.0.1	HTTP	654	GET /favicon.ico HTTP/1.1
10	127.0.0.1	HTTP	654	GET /favicon.ico HTTP/1.1

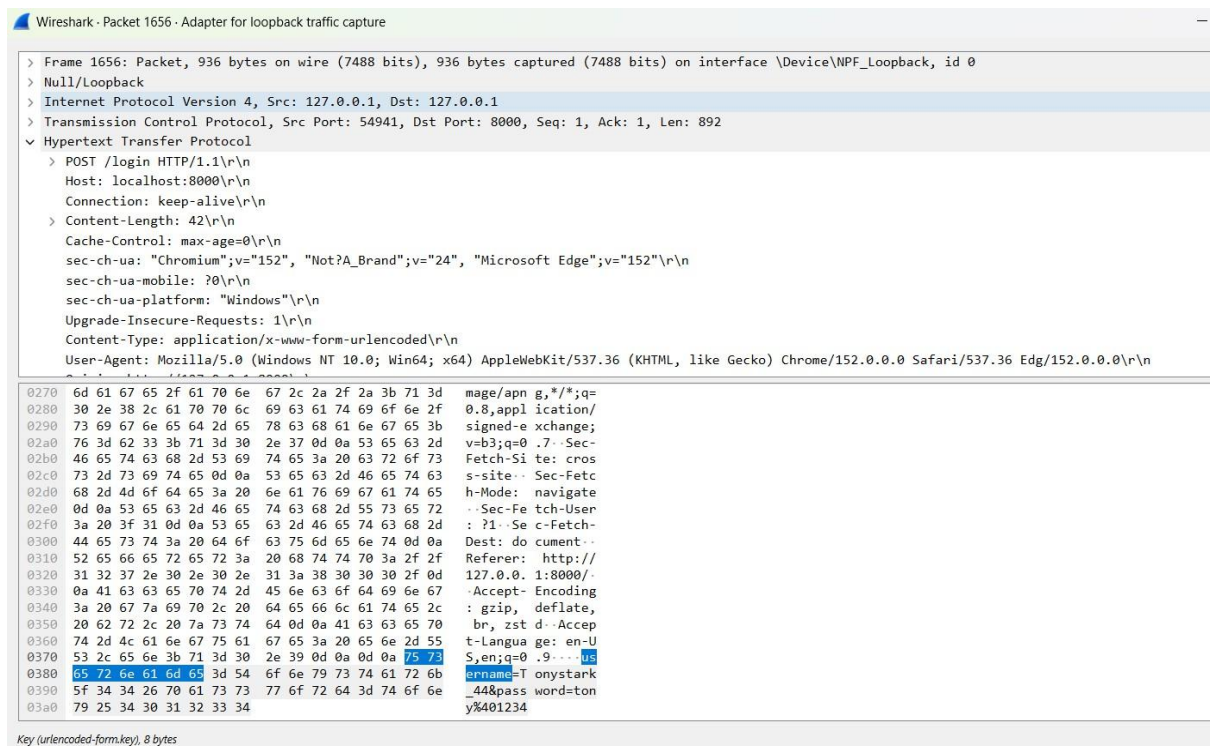
> Frame 707: Packet, 756 bytes on wire (6048 bits), 756 bytes captured (6048 bits) on interface \Device\NPF...	0000 02 00 00 00 45 00 02 f0 27 e4 40 00 80 06 d2 21 ...E-
> Null/Loopback	0010 7f 00 00 01 7f 00 00 01 ef 2f 1f 40 3f 68 4d bd ...
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	0020 b2 b7 4d ad 50 18 00 ff 4d 9a 00 00 47 45 54 20 ...M.P.
> Transmission Control Protocol, Src Port: 61231, Dst Port: 8080, Seq: 1, Ack: 1, Len: 712	0030 2f 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 .../ HTTP
> Hypertext Transfer Protocol	0040 3a 20 31 32 37 2e 30 2e 30 2e 31 3a 38 30 30 30 ...: 127.0.0.1
	0050 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 ...Conn
	0060 65 70 2d 61 6c 69 76 65 0d 0a 73 65 63 2d 63 68 ...ep-ali
	0070 2d 75 61 3a 20 22 43 68 72 6f 6d 69 75 6d 22 3b ...-ua: "I
	0080 76 3d 22 31 35 32 22 2c 20 22 4e 6f 74 3f 41 5f ...v="152'
	0090 42 72 61 6e 64 22 3b 76 3d 22 32 34 22 2c 20 22 ...Brand"
	00a0 4d 69 63 72 6f 73 6f 66 74 20 45 64 67 65 22 3b ...Microso
	00b0 76 3d 22 31 35 32 22 0d 0a 73 65 63 2d 63 68 2d ...v="152'
	00c0 75 61 2d 6d 6f 62 69 6c 65 3a 20 3f 30 0d 0a 73 ...ua-mob:
	00d0 65 63 2d 63 68 2d 75 61 2d 70 6c 61 74 66 6f 72 ...ec-ch-i

Step 5: So there are some HTTP packets are captured but we specifically looking for form data that the user submitted to the website. for that, we have a separate filter

As we know that there are main two methods used for submitting form data from web pages like login forms to the server. the methods are-

- GET
- POST

Step 6: So firstly for knowing the credential we use the first method and apply the filter for the GET methods as you can see below.
`http.request.method == "GET"`



The image shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows a packet of 936 bytes on the wire. The packet details pane on the right shows the structure of the packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The Hypertext Transfer Protocol section shows a GET request for the path `/login` on the host `localhost:8000`. The request includes several headers: `Connection: keep-alive`, `Content-Length: 42`, `Cache-Control: max-age=0`, `sec-ch-ua: "Chromium";v="152", "Not?A_Brand";v="24", "Microsoft Edge";v="152"`, `sec-ch-ua-mobile: ?0`, `sec-ch-ua-platform: "Windows"`, `Upgrade-Insecure-Requests: 1`, `Content-Type: application/x-www-form-urlencoded`, and `User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36 Edg/152.0.0.0`. The packet bytes pane at the bottom shows the raw data of the packet, which is a GET request with a `Content-Type: application/x-www-form-urlencoded` header and a `Content-Length: 42` header.

GET method

As you can see in the image there are two packets where the login page was requested with a GET request as well, but there is no form data submitted with a GET request.

Step 7: Now after checking the GET method if we didn't find the form data, then we will try the POST method for that we will apply the filter on Wireshark as you can see.
`http.request.method == "POST"`

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Wireshark successfully captured and filtered HTTP packets to identify the login form data transmitted during the experiment.

The username and password fields were observed in the HTTP POST request, demonstrating the security risk of transmitting credentials over unencrypted HTTP.

Ex. No. 4	Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)
Date	

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**

- **The receiving server's hostname/IP address.**
- **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.

- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Output Screenshots:

Original Message

Message ID	<f9a50498-8499-44d4-91f7-eca0f8cea891@dfw1s10mta1942.xt.local>
Created at:	Sun, Aug 9, 2026 at 9:50 PM (Delivered after 1 second)
From:	edX Team <news@sfmc.edx.org>
To:	dhaniya2407@gmail.com
Subject:	Celebrate IBM PC Day by learning what comes next
SPF:	PASS with IP 13.111.127.7 Learn more
DKIM:	'PASS' with domain sfmc.edx.org Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

Help

#	Delay	From ⁺	To ⁺	Protocol	Time received
0	1 sec	mta2.sfmtc.edx.org.	→ [Google] mx.google.com	ESMTPS	8/9/2026, 9:50:12 PM GMT+5:30
1			→ [Google] 2002:a05:6870:880d:b0:448:3eb2:8c8c	SMTP	8/9/2026, 9:50:12 PM GMT+5:30
2			→ [Google] 2002:a05:7010:8b89:b0:534:39cf:de51	SMTP	8/9/2026, 9:50:12 PM GMT+5:30

Whois IP 143.244.91.119

Updated 1 second

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      143.244.80.0 - 143.244.95.255
CIDR:          143.244.80.0/20
NetName:       HUBSP-8
NetHandle:     NET-143-244-80-0-1
Parent:        NET143 (NET-143-0-0-0-0)
NetType:       Direct Allocation
OriginAS:
Organization:  HubSpot, Inc. (HUBSP-8)
RegDate:       2019-12-27
Updated:       2021-04-21
Comment:       -----BEGIN CERTIFICATE-----MIIDSzCCApugAwIBAgIUTwbH/4z+Ax6hsCUGcjxxQrKOIOgwdQYJl
Ref:           https://rdap.arin.net/registry/ip/143.244.80.0

OrgName:       HubSpot, Inc.
OrgId:         HUBSP-8
Address:       2 Canal Park
City:          Cambridge
StateProv:     MA
PostalCode:    02141
Country:       US
RegDate:       2018-10-08
```

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The email header was successfully analyzed using Mail Header Analyzer (MHA), including Received, SPF, DKIM, and DMARC fields.

The analysis helped identify inconsistencies in sender information and authentication results that can indicate possible email spoofing.

Ex.No.5	Use Autopsy to create a case and import evidence
Date	

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link:

https://drive.google.com/drive/u/1/folders/1lSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Download files: [4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02](#)

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.

3. Adding a Data Source

- **Choose the Type of Data Source:**
 - After creating a case, you'll be prompted to add a data source.
 - You can add disk images, directories, logical files, or local disks.
- **Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1lSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- **Configure Ingest Modules:**

- Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- **File Analysis:**
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**
 - Compare file hashes with known databases to identify known good or bad files.

6. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.

- Save or print the report for use in your case.

7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

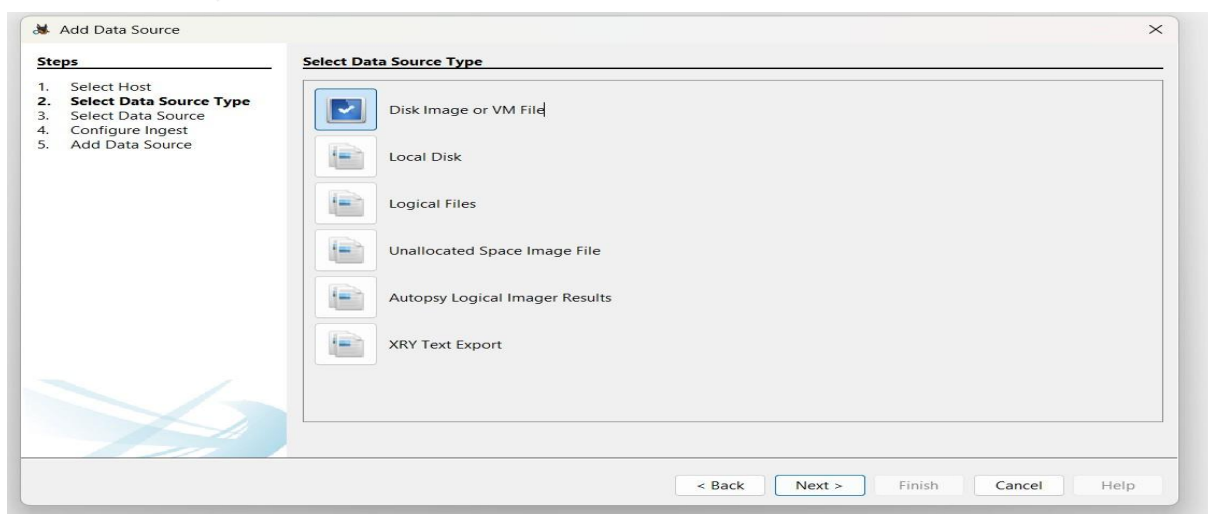
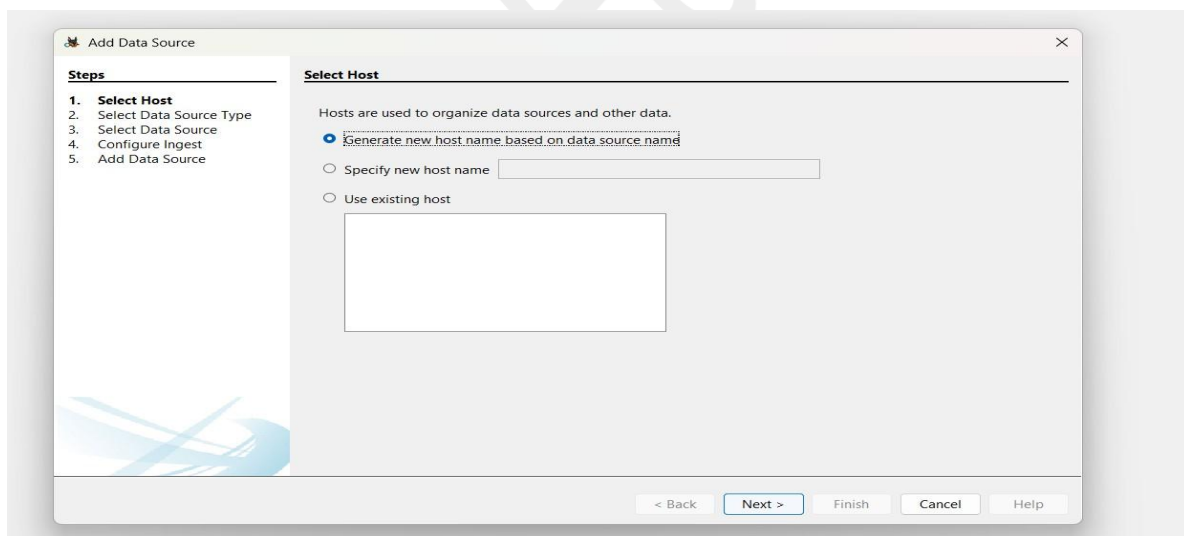
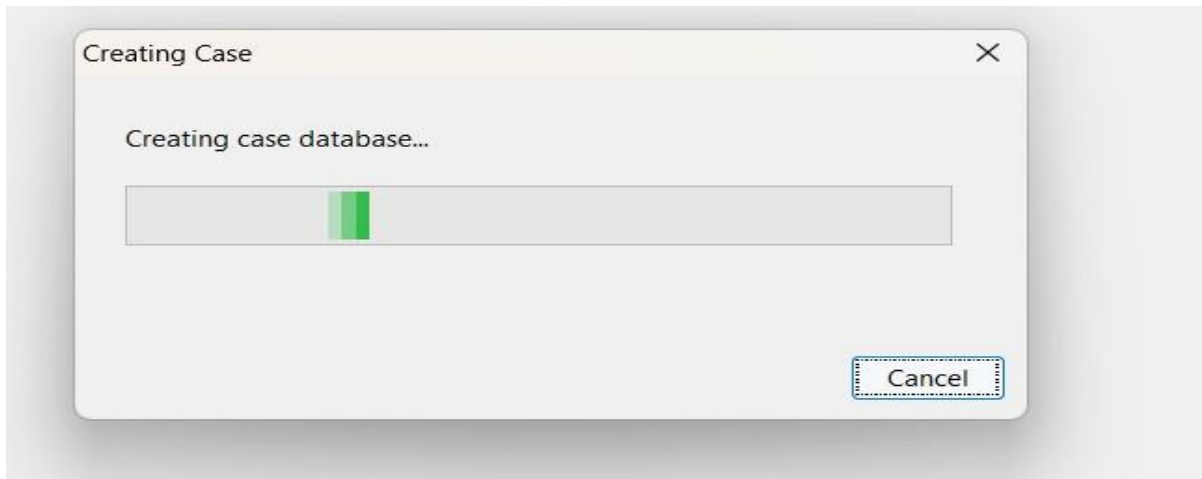
- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment

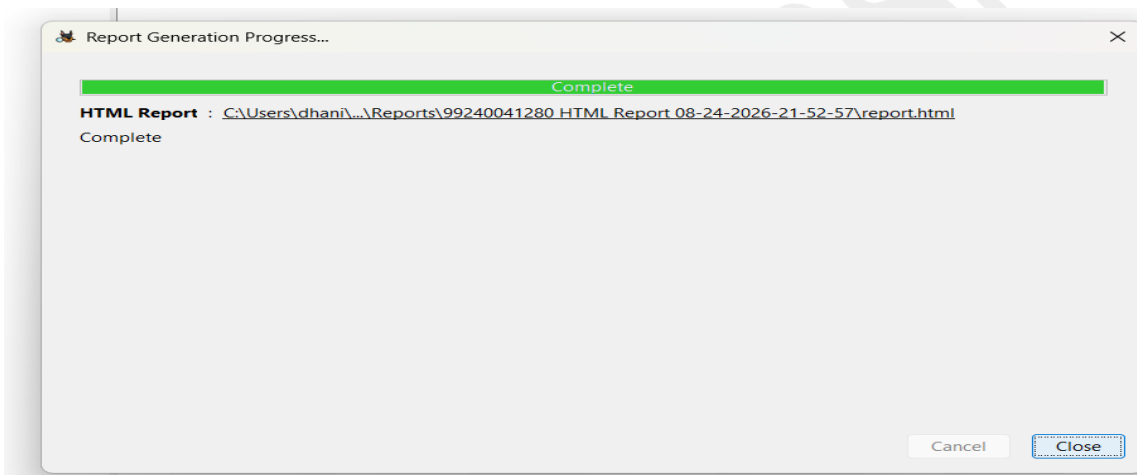
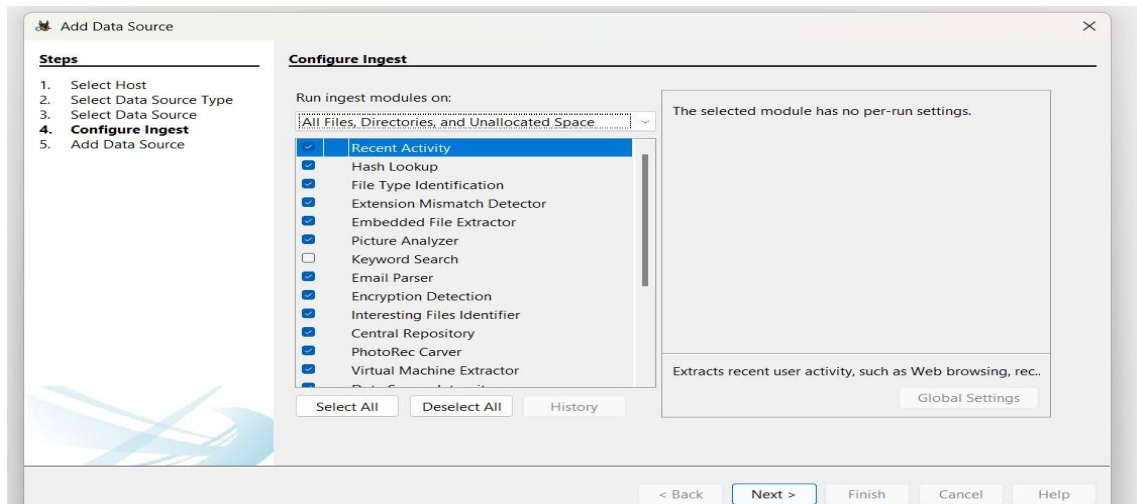
Output Screenshots:

The screenshot displays the 'New Case Information' window in the Autopsy application. The window is titled 'New Case Information' and has a close button (X) in the top right corner. On the left side, there is a 'Steps' panel with two items: '1. Case Information' and '2. Optional Information'. The main area is titled 'Case Information' and contains the following fields and controls:

- Case Name:** A text input field containing '99240041280'.
- Base Directory:** A text input field containing 'E:\', followed by a 'Browse' button.
- Case Type:** Two radio buttons: 'Single-User' (selected) and 'Multi-User'.
- Case data will be stored in the following directory:** A text input field containing 'E:\99240041280'.

At the bottom of the window, there are five buttons: '< Back', 'Next >' (highlighted in blue), 'Finish', 'Cancel', and 'Help'.





Report Navigation

Case Summary

Data Source Usage (1)

Encryption Suspected (1)

Extension Mismatch Detected (8)

Interesting Items (4)

Operating System Information (1)

Recent Documents (8)

Run Programs (59)

Tagged Files (0)

Tagged Images (0)

Tagged Results (0)

Web Bookmarks (6)

Web Categories (3)

Web Cookies (24)

Web History (18)

Autopsy Forensic Report

Warning, this report was run before ingest services completed!

HTML Report Generated on 2020/08/24 21:53:41

Case: 99240041280

Case Number: 99240041280

Number of data sources in case: 1

Examiner: Dhaniya christy J

Image Information:

4Dell Latitude CPi.E01

Timezone: Asia/Calcutta

Path: C:\Users\dhani\Downloads\4Dell Latitude CPi.E01

Software Information:

Autopsy Version: 4.23.1

Android Analyzer Module: 4.23.1

Android Analyzer (aLEAPP) Module: 4.23.1

Central Repository Module: 4.23.1

DJI Drone Analyzer Module: 4.23.1

Data Source Integrity Module: 4.23.1

Email Parser Module: 4.23.1

Embedded File Extractor Module: 4.23.1

Encryption Detection Module: 4.23.1

Extension Mismatch Detector Module: 4.23.1

File Type Identification Module: 4.23.1

GPX Parser Module: 1.2

Hash Lookup Module: 4.23.1

Interesting Files Identifier Module: 4.23.1

PhotoRec Carver Module: 7.0

Picture Analyzer Module: 4.23.1

Recent Activity Module: 4.23.1

Virtual Machine Extractor Module: 4.23.1

YARA Analyzer Module: 4.23.1

iOS Analyzer (iLEAPP) Module: 4.23.1

Ingest History:

KARE|SoC|CSE|2026-2027|Digital Forensics|Lab|Ex.No.5

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

A forensic case was successfully created in Autopsy and the E01 evidence image was imported for analysis.

The evidence was processed using ingest modules, enabling examination of files, artifacts, keywords, timelines, and other forensic data.

Ex.No.6	Use Sleuth Kit to analyze digital evidence
Date	

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

css

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Output Screenshots:

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>dir mmls.exe
Volume in drive C is Windows-SSD
Volume Serial Number is 0A36-EC13

Directory of C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin

21-09-2026  19:39                75,264 mmls.exe
             1 File(s)                75,264 bytes
             0 Dir(s) 183,371,120,640 bytes free

C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>mmls.exe
Missing image name
Usage: mmls.exe [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-BrvV] [-aAmM] [-t vstype] image [images]
        -t vstype: The type of volume system (use '-t list' for list of supported types)
        -i imgtype: The format of the image file (use '-i list' for list supported types)
        -b dev_sector_size: The size (in bytes) of the device sectors
        -o imgoffset: Offset to the start of the volume that contains the partition system (in sectors)
        -B: print the rounded length in bytes
        -r: recurse and look for other partition tables in partitions (DOS Only)
        -v: verbose output
        -V: print the version
Unless any of these are specified, all volume types are shown
        -a: Show allocated volumes
        -A: Show unallocated volumes
        -m: Show metadata volumes
        -M: Hide metadata volumes

C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>
```

```
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5
```

CONTENT INFORMATION

```
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)       Size: 8-8      Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$LOGGED_UTILITIY_STREAM (256) Size: 0-65536   Flags: Non-resident
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>fls.exe -o 63 -r -d "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01"
r/- * 0: Documents and Settings/Default User/MPC7A4.tmp
r/- * 0: Documents and Settings/Default User/MPC7A5.tmp
r/- * 6366: Documents and Settings/Default User/MPC7A6.tmp
r/- * 6367: Documents and Settings/Default User/MPC7A7.tmp
r/- * 6368: Documents and Settings/Default User/MPC7A8.tmp
r/- * 6369: Documents and Settings/Default User/MPC7A9.tmp
r/- * 6370: Documents and Settings/Default User/MPC7AA.tmp
r/- * 6371: Documents and Settings/Default User/MPC7AB.tmp
r/- * 6372: Documents and Settings/Default User/MPC7AC.tmp
r/- * 0: Documents and Settings/Default User/MPC7B8.tmp
r/- * 0: Documents and Settings/Default User/MPC7B9.tmp
r/- * 6386: Documents and Settings/Default User/MPC7BA.tmp
r/- * 6387: Documents and Settings/Default User/MPC7BB.tmp
r/- * 6388: Documents and Settings/Default User/MPC7BC.tmp
r/- * 6389: Documents and Settings/Default User/MPC7BD.tmp
r/- * 6390: Documents and Settings/Default User/MPC7BE.tmp
r/- * 6391: Documents and Settings/Default User/MPC7BF.tmp
r/- * 6392: Documents and Settings/Default User/MPC7C0.tmp
r/- * 0: Documents and Settings/Default User/MPC7CC.tmp
r/- * 0: Documents and Settings/Default User/MPC7CE.tmp
r/- * 6408: Documents and Settings/Default User/MPC7CF.tmp
r/- * 6409: Documents and Settings/Default User/MPC7D0.tmp
r/- * 6410: Documents and Settings/Default User/MPC7D1.tmp
r/- * 6411: Documents and Settings/Default User/MPC7D2.tmp
r/- * 6412: Documents and Settings/Default User/MPC7D3.tmp
r/- * 6413: Documents and Settings/Default User/MPC7D4.tmp
r/- * 0: Documents and Settings/Default User/MPC7E0.tmp
r/- * 0: Documents and Settings/Default User/MPC7E2.tmp
r/- * 6428: Documents and Settings/Default User/MPC7E3.tmp
r/- * 6429: Documents and Settings/Default User/MPC7E4.tmp
r/- * 6430: Documents and Settings/Default User/MPC7E5.tmp
r/- * 6431: Documents and Settings/Default User/MPC7E6.tmp
r/- * 6432: Documents and Settings/Default User/MPC7E7.tmp
r/- * 6433: Documents and Settings/Default User/MPC7E8.tmp
r/- * 0: Documents and Settings/Default User/MPC7F4.tmp
r/- * 0: Documents and Settings/Default User/MPC7F6.tmp
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>icat.exe -o 63 "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01" 3726 > extracted_h323log.txt
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>dir extracted_h323log.txt
Volume in drive C is Windows-SSD
Volume Serial Number is 0A36-EC13
```

```
Directory of C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin
```

```
21-09-2026 20:00          0 extracted_h323log.txt
                1 File(s)          0 bytes
                0 Dir(s) 183,364,120,576 bytes free
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>notepad extracted_h323log.txt
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>istat.exe -o 63 "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01" 3726-128-1
MFT Entry Header Values:
Entry: 3726          Sequence: 2
$LogFile Sequence Number: 11115366
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ()
Created: 2004-08-19 22:47:33.514881600 (India Standard Time)
File Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
MFT Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
Accessed: 2004-08-19 22:47:33.514881600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: h323log.txt
Parent MFT Entry: 459 Sequence: 459
Allocated Size: 0 Actual Size: 0
Created: 2004-08-19 22:47:33.514881600 (India Standard Time)
File Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
MFT Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
Accessed: 2004-08-19 22:47:33.514881600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 88
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 148
Type: $DATA (128-1) Name: N/A Resident size: 0
```

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The Sleuth Kit successfully analyzed the forensic disk image and identified file system and deleted file information. The metadata of the selected deleted file was examined using the istat command.

Ex.No.7	Use AF Logical OSE to extract data from an Android device
Date	

Description

AF Logical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AF Logical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

1. Download AF Logical OSE:

- Visit the [AF Logical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AF Logical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:
Bash

adb devices

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AF Logical OSE

1. Run AF Logical OSE:

- Navigate to the directory where you have downloaded or extracted AF Logical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AF Logical OSE APK on the device:
Bash
adb install af.logical.apk
- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AF Logical OSE App on the Device:

- On the Android device, open the AF Logical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named af logical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:
bash
adb pull /sdcard/aflogical /path/to/destination
- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

1. Uninstall AF Logical OSE:

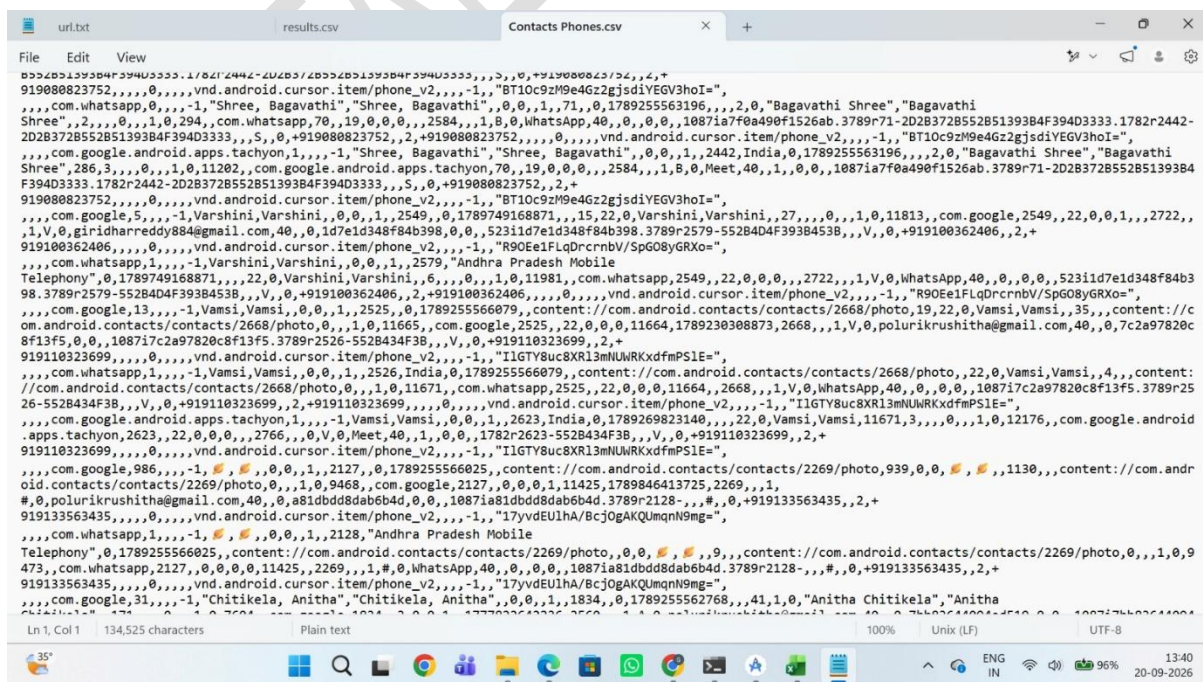
- Once data extraction is complete, you can uninstall the AF Logical OSE app from the Android device by running:

adb uninstall com.viaforensics.android.aflogical

- Safely disconnect the Android device from your computer.

By following these steps, you can use AF Logical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Output Screenshots:



```

C:\Users\user\Downloads\platform-tools-latest-windows>cd platform-tools

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb devices
List of devices attached
10BE6X06HH000ED device

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb install -r "C:\Users\user\AndroidStudioProjects\AFLogicalOSE
2\app\build\outputs\apk\debug\app-debug.apk"
Performing Streamed Install
Success

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb shell ls /sdcard/Android/data/com.viaforensics.android.aflog
ical_ose/files/forensics
20260920.1240
20260920.1242

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb shell ls -R /sdcard/Android/data/com.viaforensics.android.af
logical_ose/files/forensics
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics:
20260920.1240
20260920.1242

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1240:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1242:
info.xml

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb logcat -c

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb logcat -d -t 500 | findstr /I "AndroidRuntime FATAL EXCEPTIO
N SecurityException IllegalArgumentException viaforensics aflogical"

```

```

C:\Windows\System32\cmd X + v
logical_ose/files/forensics
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics:
20260920.1240
20260920.1242
20260920.1248
20260920.1300
20260920.1332

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1240:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1242:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1248:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1300:
info.xml

/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files/forensics/20260920.1332:
Contacts Phones.csv
info.xml

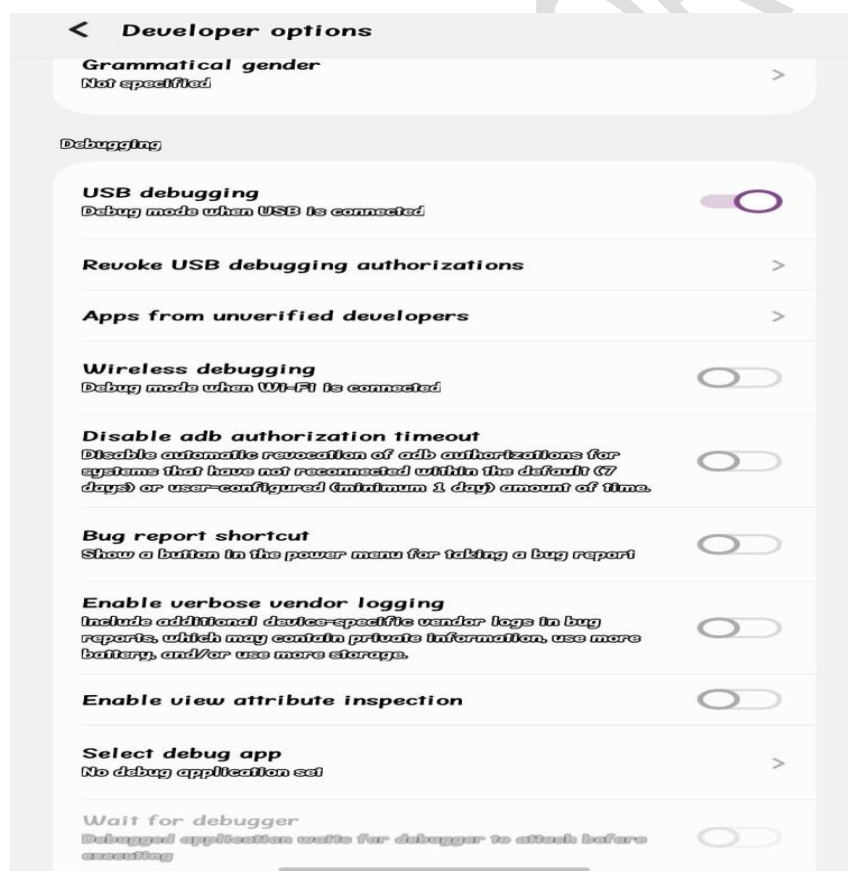
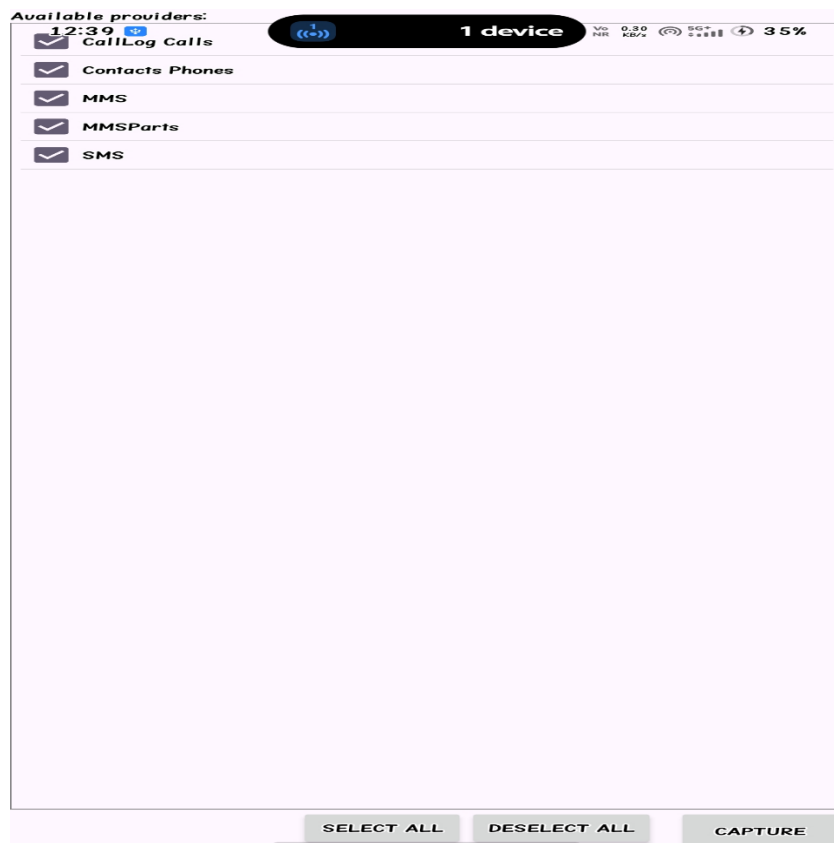
C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>if not exist C:\AFLogical_Result mkdir C:\AFLogical_Result

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>adb pull /sdcard/Android/data/com.viaforensics.android.aflogical
_ose/files/forensics/ C:\AFLogical_Result
/sdcard/Android/data/com.viaforensics.android.aflogical_ose/files...cs/: 6 files pulled, 0 skipped. 5.8 MB/s (288084 bytes in 0.047s)

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>explorer C:\AFLogical_Result

C:\Users\user\Downloads\platform-tools-latest-windows\platform-tools>

```



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

AF Logical OSE was used to perform logical data extraction from an Android device. The extracted data was transferred to the computer and analyzed in CSV format for forensic investigation.

Ex. No. 8	Use Steg-Expose to detect hidden data in images
Date	

Description

Steg Expose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded. Below is a step-by-step guide on how to use Steg Expose to detect hidden data in images:

Prerequisites:

1. **Java Runtime Environment (JRE):** Steg Expose is a Java-based tool, so ensure you have JRE installed.
2. **Steg Expose Tool:** Download Steg Expose from the [official GitHub repository](#).

Step-by-Step Process:

1. Download and Set Up Steg Expose:

- **Download the tool:** Go to [Steg Expose GitHub](#) and download the .jar file.
- **Install Java:** Ensure that Java is installed on your machine. You can download it from the official site.
- **Prepare environment:** Place the .jar file in a folder where you plan to work on the image files.

2. Select Images for Analysis:

- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.

3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS).

4. Run Steg Expose on an Image:

- Use the following command to check a single image for hidden data:

bash

```
java -jar StegExpose.jar <image_file_path>
```

Example:

bash

```
java -jar StegExpose.jar test_image.png
```

- Replace <image_file_path> with the actual path of the image you want to check.

5. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- **Output Explanation:**
 - StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.
 - **Threshold values** (suggested by the tool):
 - **Less than 0.2:** Image is clean (no hidden data).
 - **0.2 - 0.3:** Possibly some hidden data.
 - **Above 0.3:** Likely that steganography is present.

6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder. For example:

```
bash
```

```
java -jar StegExpose.jar <folder_path>
```

- Replace <folder_path> with the path to the folder containing the images.

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

```
bash
```

```
java -jar StegExpose.jar --help
```

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Example:

```
bash
```

```
java -jar StegExpose.jar suspect_image.png
```

Output:

```
makefile
```

```
Analyzing suspect_image.png...
```

```
Result: 0.4
```

```
Steganography likely present
```

This output indicates that hidden data is likely present in the image suspect_image.png.

Output Screenshots:

```

Command Prompt
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\dhani> java -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)
PS C:\Users\dhani>

```

2	File name	Above steg	Secret mes	Primary Se	Chi Square	Sample Pa	RS analysis	Fusion (mean)
3	image 1.pr	FALSE	1370	0.003522	0	null	0.016585	0.006703
4	image 2.pr	FALSE	5271	0.050289	1.65E-14	null	0.02748	0.025923
5	image 3.pr	FALSE	863	0.012388	0	0.00537	0.016264	0.008506
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								

```

PS C:\StegExposeLab> Start-Process .\results.csv
PS C:\StegExposeLab> Get-Content .\results.csv

File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
image 1.png,false,1370,0.003522431921441729,0.0,null,0.016585235152236963,0.0067025556912262306
image 2.png,false,5271,0.050289071306961974,1.649152512663717E-14,null,0.027479554543971343,0.02592287528364994
image 3.png,false,863,0.012387543663056765,0.0,0.0053704199757471675,0.016264357216348613,0.008505580213788135
PS C:\StegExposeLab>

```

```
PS C:\Users\dhani> cd C:\StegExposeLab
PS C:\StegExposeLab> Get-ChildItem

Directory: C:\StegExposeLab
```

Mode	LastWriteTime		Length	Name
d-----	19-09-2026	16:30		images
-a----	19-09-2026	16:21	3060550	StegExpose.jar

```
PS C:\StegExposeLab> Get-ChildItem .\images
```

```
Directory: C:\StegExposeLab\images
```

Mode	LastWriteTime		Length	Name
-a----	19-09-2026	16:30	613380	image 1.png
-a----	19-09-2026	16:30	610003	image 2.png
-a----	19-09-2026	16:30	304330	image 3.png

```
PS C:\StegExposeLab> java -jar .\StegExpose.jar .\images
PS C:\StegExposeLab> java -jar .\StegExpose.jar .\images default default results.csv
PS C:\StegExposeLab> Get-ChildItem
```

```
Directory: C:\StegExposeLab
```

Mode	LastWriteTime		Length	Name
d-----	19-09-2026	16:30		images
-a----	19-09-2026	16:34	475	results.csv
-a----	19-09-2026	16:21	3060550	StegExpose.jar

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Steg Expose successfully analyzed the image and generated a steganography detection score. The result indicated whether hidden data was likely present, demonstrating the use of Steg Expose for image-based forensic analysis.

Ex. No. 9	Use Process Explorer to identify suspicious processes
Date	

Description:

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications.

Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Project Steps:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program:

- Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."

Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. Process Tree:

- The main window shows a tree of all running processes, with each child process listed under its parent.
- Processes are color-coded to indicate their status:
 - Pink:** Suspended processes.
 - Light Blue:** Processes running under the same user as you.
 - Dark Blue:** Services or processes running under system accounts.

- **Green:** New processes (appearing and disappearing quickly may be suspicious).
- **Red:** Processes that have just exited.

2. Columns:

- Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:
 - Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.

Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.

Step 5: Kill or Suspend Suspicious Processes

If you find a process that you believe is malicious:

1. Kill the Process:

- Right-click the suspicious process and select Kill Process to terminate it immediately.

2. Suspend the Process:

- If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.

3. Remove the Source File:

- Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
- Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.

Step 6: Scan Your System

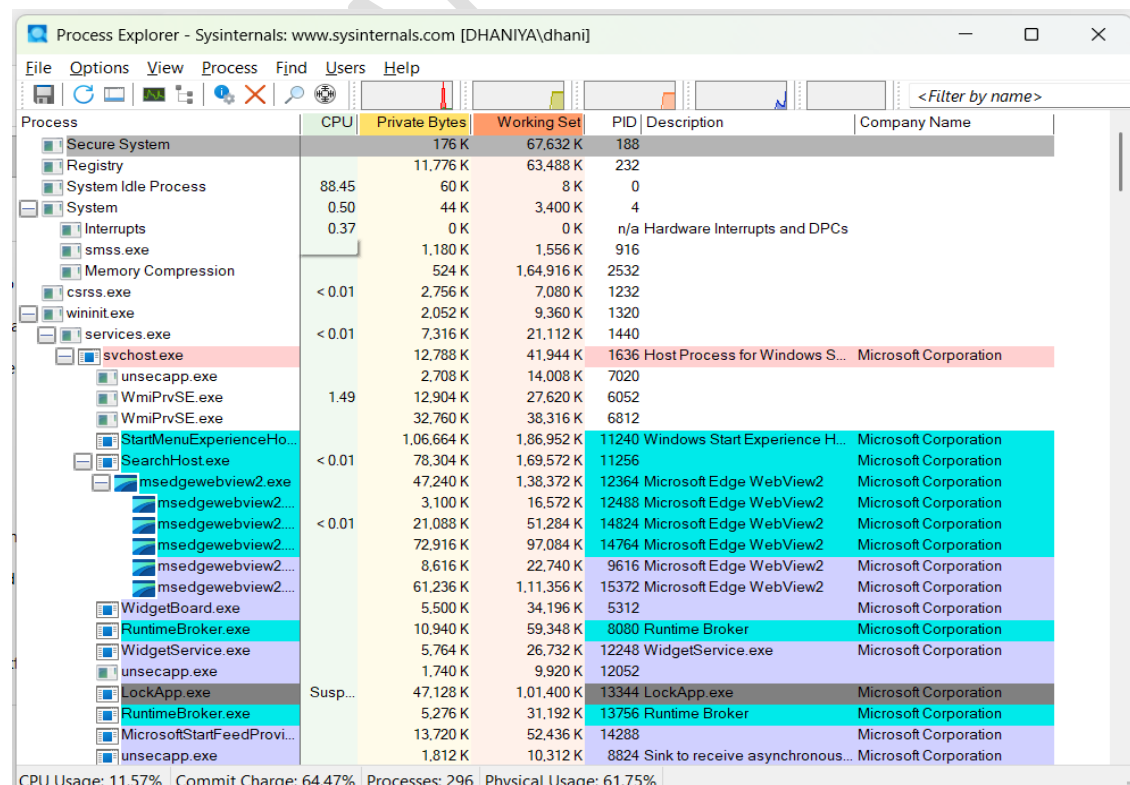
After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.

Example of Identifying a Suspicious Process

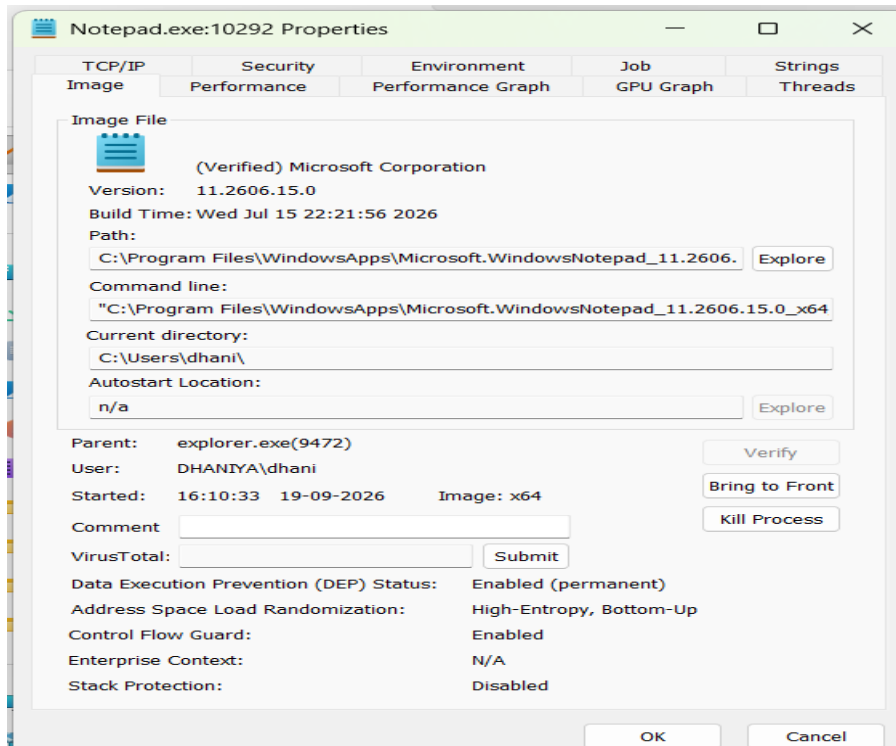
1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.
5. You kill the process and remove the executable file from the system.

Output Screenshots:



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		176 K	67,632 K	188		
Registry		11,776 K	63,488 K	232		
System Idle Process	88.45	60 K	8 K	0		
System	0.50	44 K	3,400 K	4		
Interrupts	0.37	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,180 K	1,556 K	916		
Memory Compression		524 K	1,64,916 K	2532		
csrss.exe	< 0.01	2,756 K	7,080 K	1232		
wininit.exe		2,052 K	9,360 K	1320		
services.exe	< 0.01	7,316 K	21,112 K	1440		
svchost.exe		12,788 K	41,944 K	1636	Host Process for Windows S...	Microsoft Corporation
unsecapp.exe		2,708 K	14,008 K	7020		
WmiPrvSE.exe	1.49	12,904 K	27,620 K	6052		
WmiPrvSE.exe		32,760 K	38,316 K	6812		
StartMenuExperienceHo...		1,06,664 K	1,86,952 K	11240	Windows Start Experience H...	Microsoft Corporation
SearchHost.exe	< 0.01	78,304 K	1,69,572 K	11256		Microsoft Corporation
msedgewebview2.exe		47,240 K	1,38,372 K	12364	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		3,100 K	16,572 K	12488	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...	< 0.01	21,088 K	51,284 K	14824	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		72,916 K	97,084 K	14764	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		8,616 K	22,740 K	9616	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2...		61,236 K	1,11,356 K	15372	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		5,500 K	34,196 K	5312		Microsoft Corporation
RuntimeBroker.exe		10,940 K	59,348 K	8080	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,764 K	26,732 K	12248	WidgetService.exe	Microsoft Corporation
unsecapp.exe		1,740 K	9,920 K	12052		
LockApp.exe	Susp...	47,128 K	1,01,400 K	13344	LockApp.exe	Microsoft Corporation
RuntimeBroker.exe		5,276 K	31,192 K	13756	Runtime Broker	Microsoft Corporation
MicrosoftStartFeedProvi...		13,720 K	52,436 K	14288		Microsoft Corporation
unsecapp.exe		1,812 K	10,312 K	8824	Sink to receive asynchronous...	Microsoft Corporation

CPU Usage: 11.57% | Commit Charge: 64.47% | Processes: 296 | Physical Usage: 61.75%



Process Explorer - Sysinternals: www.sysinternals.com [DHANIYA\dhani]

File Options View Process Find Users Help

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
msedge.exe		66,864 K	1,28,212 K	23488	Microsoft Edge	Microsoft Corporation
msedge.exe		9,408 K	29,248 K	18792	Microsoft Edge	Microsoft Corporation
msedge.exe	0.37	3,97,748 K	4,43,960 K	6020	Microsoft Edge	Microsoft Corporation
msedge.exe	< 0.01	81,104 K	1,40,920 K	27520	Microsoft Edge	Microsoft Corporation
msedge.exe		1,32,292 K	1,94,228 K	27568	Microsoft Edge	Microsoft Corporation
msedge.exe		8,052 K	17,952 K	26956	Microsoft Edge	Microsoft Corporation
msedge.exe		31,200 K	74,416 K	26212	Microsoft Edge	Microsoft Corporation
msedge.exe		18,308 K	34,516 K	20004	Microsoft Edge	Microsoft Corporation
MathWorksServiceHost.exe	< 0.01	78,884 K	1,77,808 K	24732	MathWorks Service Host	The MathWorks Inc.
MathWorksServiceHost-M...	< 0.01	25,652 K	60,340 K	25296	MathWorks Service Host	The MathWorks Inc.
mscopilot.exe		45,820 K	1,41,124 K	25496	Copilot	Microsoft Corporation
mscopilot.exe		2,888 K	12,928 K	25536	Copilot	Microsoft Corporation
mscopilot.exe		62,964 K	48,872 K	17956	Copilot	Microsoft Corporation
mscopilot.exe		20,380 K	47,548 K	18120	Copilot	Microsoft Corporation
mscopilot.exe		8,148 K	18,764 K	23420	Copilot	Microsoft Corporation
ONENOTE.MEXE		15,296 K	9,100 K	23460	Send to OneNote Tool	Microsoft Corporation
procxp64.exe	0.12	40,852 K	75,268 K	24332	Sysinternals Process Explorer	Sysinternals - www.sysinter...
Notepad.exe	0.12	1,21,952 K	1,85,980 K	10292		
SnippingTool.exe	1.49	54,172 K	95,112 K	3028		
BraveCrashHandler.exe		1,800 K	2,076 K	10976		
BraveCrashHandler64.exe		1,796 K	1,828 K	11004		
LenovoVantage.exe	< 0.01	1,86,252 K	26,996 K	17916	Lenovo Vantage	Lenovo
msedgewebview2.exe	< 0.01	55,384 K	11,856 K	18528	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,928 K	484 K	18552	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		90,012 K	1,268 K	19052	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		20,316 K	4,628 K	19064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,284 K	764 K	19196	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	82,468 K	18,712 K	19416	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		21,048 K	536 K	20068	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		19,988 K	212 K	20152	Microsoft Edge WebView2	Microsoft Corporation

Process Explorer - Sysinternals: www.sysinternals.com [DHANIYA\dhani]

File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
msedge.exe		9,400 K	29,240 K	18792	Microsoft Edge	Microsoft Corporation
msedge.exe	0.37	3,97,236 K	4,43,660 K	6020	Microsoft Edge	Microsoft Corporation
msedge.exe	< 0.01	81,140 K	1,41,040 K	27520	Microsoft Edge	Microsoft Corporation
msedge.exe		1,32,668 K	1,94,180 K	27568	Microsoft Edge	Microsoft Corporation
msedge.exe		8,008 K	17,932 K	26956	Microsoft Edge	Microsoft Corporation
msedge.exe		31,168 K	74,400 K	26212	Microsoft Edge	Microsoft Corporation
msedge.exe		18,308 K	34,516 K	20004	Microsoft Edge	Microsoft Corporation
MathWorksServiceHost.exe	0.12	78,844 K	1,77,780 K	24732	MathWorks Service Host	The MathWorks Inc.
MathWorksServiceHost-M...	< 0.01	25,652 K	60,340 K	25296	MathWorks Service Host	The MathWorks Inc.
mscopilot.exe		45,820 K	1,41,128 K	25496	Copilot	Microsoft Corporation
mscopilot.exe		2,888 K	12,928 K	25536	Copilot	Microsoft Corporation
mscopilot.exe		62,964 K	48,872 K	17956	Copilot	Microsoft Corporation
mscopilot.exe		20,380 K	47,548 K	18120	Copilot	Microsoft Corporation
mscopilot.exe		8,104 K	18,744 K	23420	Copilot	Microsoft Corporation
ONENOTEM.EXE		15,296 K	9,100 K	23460	Send to OneNote Tool	Microsoft Corporation
procexp64.exe	0.62	40,852 K	75,268 K	24332	Sysinternals Process Explorer	Sysinternals - www.sysinter...
Notepad.exe	< 0.01	1,21,980 K	1,88,416 K	10292		
BraveCrashHandler.exe		1,800 K	2,008 K	10976		
BraveCrashHandler64.exe		1,796 K	1,820 K	11004		
LenovoVantage.exe	< 0.01	1,85,916 K	17,692 K	17916	Lenovo Vantage	Lenovo
msedgewebview2.exe	< 0.01	55,388 K	9,976 K	18528	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,928 K	0 K	18552	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		90,032 K	19,648 K	19052	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		20,312 K	3,496 K	19064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,284 K	504 K	19196	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	82,156 K	29,904 K	19416	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		21,048 K	1,160 K	20068	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		19,988 K	1,120 K	20152	Microsoft Edge WebView2	Microsoft Corporation
SDTray.exe	0.25	12,552 K	39,152 K	25724	Spybot - Search & Destroy tr...	Safer-Networking Ltd.

CPU Usage: 3.20% Commit Charge: 64.19% Processes: 288 Physical Usage: 61.65%

chrome.exe:13056 Properties

Image Performance Performance Graph GPU Graph Threads TCP/IP Security Environment Job Strings

☒ Resolve addresses

Proto...	Local Address	R...	State
UDP	dhaniya:5353	..*	
UDP	dhaniya:5353	..*	
UDP	dhaniya:5353	..*	
UDP	dhaniya:5353	..*	
UDPv6	dhaniya:5353	..*	
UDPv6	dhaniya:5353	..*	

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Process Explorer was successfully used to monitor running processes and identify potentially suspicious processes based on process details, paths, signatures, and resource usage. The experiment demonstrated how process behavior and network activity can be examined for forensic investigation and malware detection.

Ex. No. 10	Use Ghidra to disassemble and analyze the malware code
Date	

Project Objectives

- To provide a hands-on guide to disassembling and analyzing malware with Ghidra.
- To enable readers to recognize common malware functionalities, such as persistence mechanisms, anti-analysis techniques, and network communications.
- To equip users with the skills to interpret low-level assembly code and understand high-level behaviours.

GitHub Repository Components

A GitHub repository for this project could include:

1. README.md

- Overview: Brief description of Ghidra, its installation, and how it is used for malware analysis.
- Requirements: List of requirements, including Ghidra, a VM setup, and any sample binaries (preferably benign samples or detailed steps to safely acquire controlled malware samples).
- Project Outline: A breakdown of the tutorial steps, linking to various sections or files within the repository.

2. Tutorial Steps (Markdown or Jupyter Notebooks)

- Step-by-step instructions, broken down into digestible sections, such as:
 - Environment Setup: Instructions on setting up a virtualized and isolated environment.
 - Initial Analysis: How to load a binary into Ghidra, perform auto-analysis, and identify entry points.
 - Function Analysis: Methods for identifying and analyzing critical functions, decompiling code, and using Ghidra's cross-referencing features.
 - String and Import Analysis: How to locate and interpret strings and imports related to malicious behavior.

- Advanced Techniques: Dynamic analysis, control flow graphs, custom scripting with Ghidra's Python or Java capabilities, and obfuscation detection.

3. Example Scripts (Python or Java)

- Scripts to automate repetitive analysis tasks, such as:
 - Extracting and listing all string references.
 - Identifying network-related functions or common indicators of malware.
 - Labeling functions and data segments based on known patterns (e.g., file system manipulation, registry edits).
- Script Examples: Python or Java code snippets demonstrating API usage in Ghidra (using Ghidra's Headless Analyzer, for example, for automation).

4. Sample Data (Benign Binary or Hex Dump)

- Due to safety and ethical concerns, include only safe, benign binaries or a hex dump example rather than live malware. This can allow users to learn and practice without the risk of executing malicious code.
- Sample binaries designed to exhibit typical malware-like behaviors, such as:
 - Executing system calls.
 - Writing to files.
 - Sending network data.
- Alternatively, a dummy binary generator script that creates binaries with similar structures to malware but without harmful payloads.

5. Documentation and Comments

- Include explanations of the code within each script, detailing what each function does.
- Inline comments in disassembly views, providing readers with insights into assembly commands, control flow structures, and the decompiled code.

6. Report Template

- A reporting template for documenting findings, which could be in Markdown or PDF format. The template should cover:
 - Summary of Analysis: Brief description of the suspected malware.
 - Function Analysis: Summary of the key functions, their purpose, and the findings.

- Behavioral Indicators: Summary of malware behaviors, such as persistence, file system manipulation, or network communications.
- Recommendations: Suggestions on further actions, such as containment, eradication, or further analysis.

Example Repository Structure

plaintext

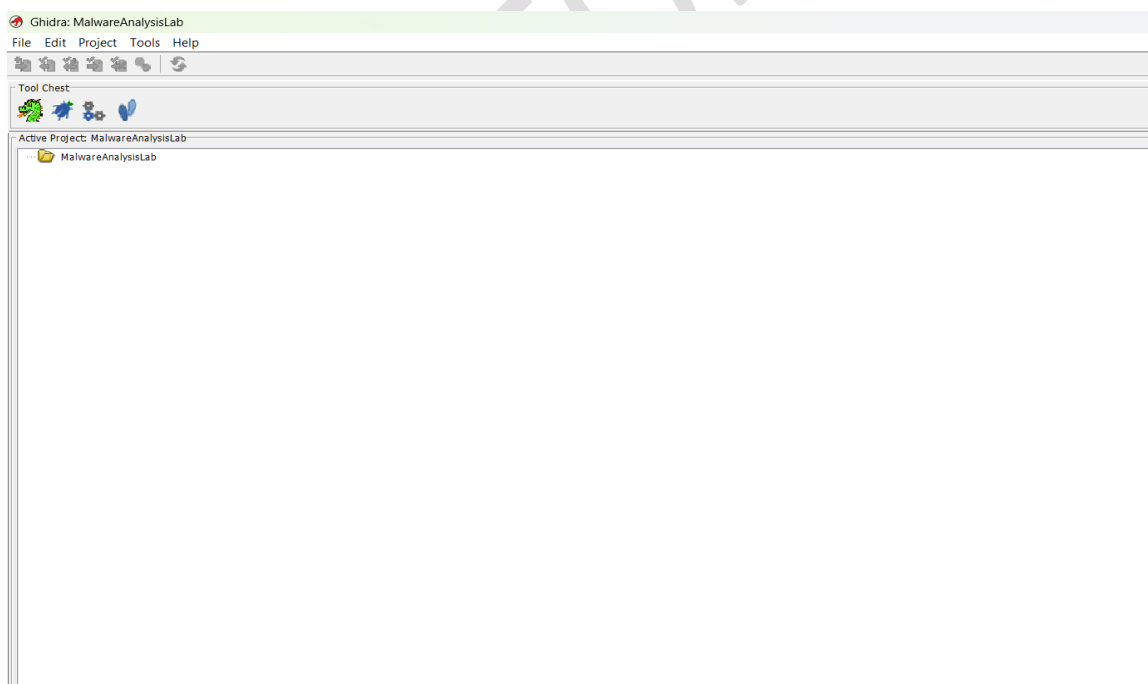
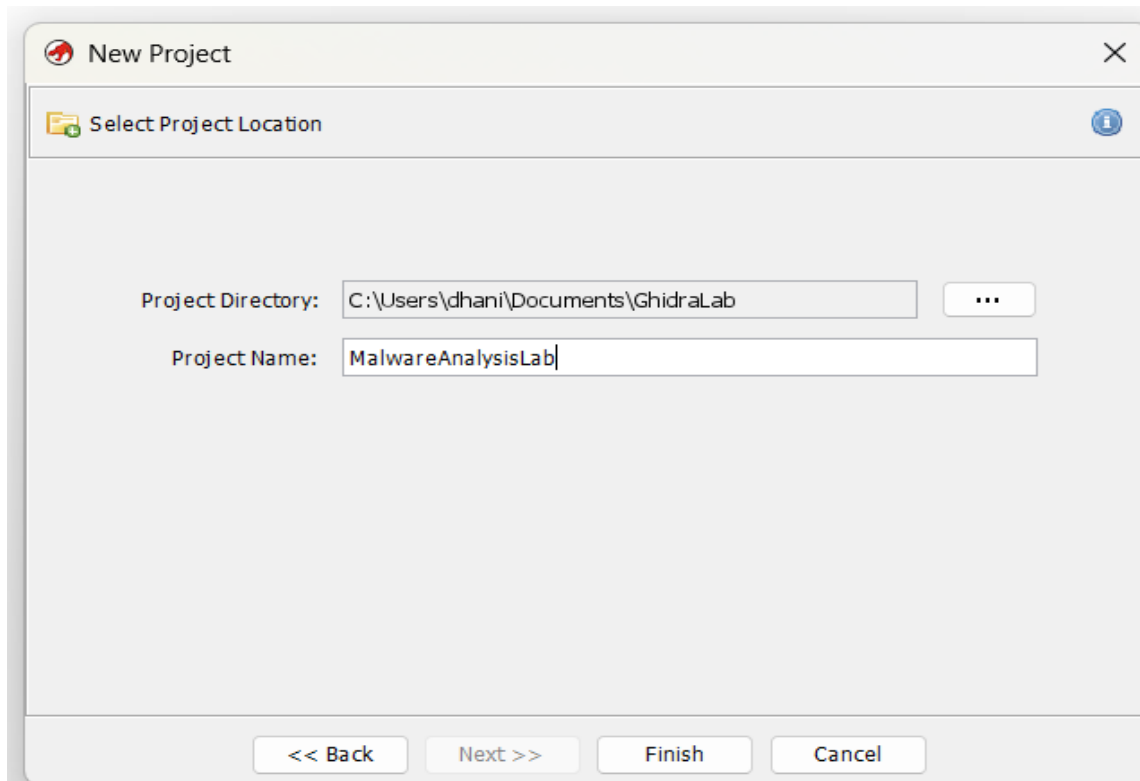
ghidra-malware-analysis-project/

```
|— README.md
|— docs/
|   |— Tutorial_Step1.md
|   |— Tutorial_Step2.md
|   └─ Tutorial_StepN.md
|— scripts/
|   |— extract_strings.py
|   |— network_analysis.py
|   └─ label_functions.py
|— samples/
|   |— benign_sample1.bin
|   └─ benign_sample2.hex
|— templates/
|   └─ analysis_report_template.md
└─ screenshots/
    |— function_graph_example.png
    └─ decompiler_example.png
```

Project Development and Future Work

- Expand Scripts: Continue adding Ghidra scripts to automate complex analysis tasks.
- Community Contributions: Encourage community members to contribute additional benign samples, analysis scripts, and documentation improvements.
- Advanced Topics: Include guides for advanced techniques, such as unpacking obfuscated code, identifying encryption routines, and using the P-Code emulator for runtime analysis.

Output Screenshots:



Defined Strings [CodeBrowser: MalwareAnalysisLab/ping.exe]

Edit Help

Defined Strings - 138 items

Location	String Value	String Representation	Data Type
140000000	MZ	"MZ"	char[2]
140000100	PE	"PE"	char[4]
140000208	.text	".text"	char[8]
140000230	.fothk	".fothk"	char[8]
140000238	.rdata	".rdata"	char[8]
140000280	.data	".data"	char[8]
1400002A8	.pdata	".pdata"	char[8]
1400002D0	.rsrc	".rsrc"	char[8]
1400002F8	.reloc	".reloc"	char[8]
1400054C8	OutputEncoding	"OutputEncoding"	unicode
1400054A8	Unicode	"Unicode"	unicode
1400054B8	Ansi	"Ansi"	unicode
140005508	UTF8	"UTF8"	unicode
140005518	UTF-8	"UTF-8"	unicode
140005540	SYSTEM\\CurrentControlSet\\Services\\Tcpip\\Parameters	"SYSTEM\\CurrentControlSet\\Services\\Tcpip\\Parameters"	unicode
1400055A8	DefaultTTL	"DefaultTTL"	unicode
1400055C0	IPv4	"IPv4"	unicode
1400055D0	IPv6	"IPv6"	unicode
140005600	RSDS	"RSDS"	ds
140005708	ping.pdb	"ping.pdb"	ds
14000582C	.gchcnt	".gchcnt"	ds
140005850	.rdata	".rdata"	ds
140005860	.rdata\$voltnid	".rdata\$voltnid"	ds
140005890	.data	".data"	ds
140005910	.pdata	".pdata"	ds
14000592	.exit	".exit"	ds
1400059A	._get_oshandle	"._get_oshandle"	ds
1400059C	._wcsicmp	"._wcsicmp"	ds
1400059E8	flush	".flush"	ds
1400059C2	svctype	".svctype"	ds
1400059C4	wcsnul	".wcsnul"	ds
1400059E8	._setmode	"._setmode"	ds
1400059E4	._write	"._write"	ds
1400059E4	._fopen	"._fopen"	ds
1400059F8	wcchr	".wcchr"	ds
140006002	fgetpos	".fgetpos"	ds
14000600C	fwprintf	".fwprintf"	ds
140006018	._XcptFilter	"._XcptFilter"	ds
140006028	._ymsg_exit	"._ymsg_exit"	ds
140006034	._wgetmainargs	"._wgetmainargs"	ds
140006046	._set_app_type	"._set_app_type"	ds
140006058	._exit	"._exit"	ds
140006060	._cexit	"._cexit"	ds
14000606A	._setusermatherr	"._setusermatherr"	ds
140006074	._initterm	"._initterm"	ds
14000608A	._C_specific_handler	"._C_specific_handler"	ds
140006092	._fmode	"._fmode"	ds
14000609C	._commode	"._commode"	ds

Filter:

CodeBrowser: MalwareAnalysisLab/ping.exe

File Edit Analysis BSim Graph Navigation Search Select Tools Window Help

Program Trees

- ping.exe
 - Headers
 - .text
 - .fothk
 - .rdata
 - .data
 - .pdata
 - .rsrc
 - .reloc
 - Debug Data
 - tdb

Program Tree X

Symbol Tree

- API-MS-WIN-CORE-PROCESS-THREADS-L1-1-0.DLL
- API-MS-WIN-CORE-PROFILE-L1-1-0.DLL
- API-MS-WIN-CORE-REGISTRY-L1-1-0.DLL
- API-MS-WIN-CORE-RTLSUPPORT-L1-1-0.DLL
- API-MS-WIN-CORE-STRING-L1-1-0.DLL
- API-MS-WIN-CORE-SYNCH-L1-2-0.DLL
- API-MS-WIN-CORE-SYINFO-L1-1-0.DLL
- IPHLPAPI.DLL
- MSVCRT.DLL
- NTDLL.DLL
- WS2_32.DLL
- Exports

Filter:



```

1400010f0 - FUN_1400010f0
undefined __fastcall FUN_1400010f0(void)
undefined <UNASSIGNED> <RETURN>
undefined8 Stack[-0x18]:8 local_18
FUN_1400010f0
...10f0 SUB RSP,0x38
...10f4 MOV EAX,dword ptr [DAT_1400076...]
...10fa LEA RS,[DAT_140007158]
...1101 MOV RSD,dword ptr [DAT_1400076...]
...1108 LEA RDX,[DAT_140007150]
...110f MOV dword ptr [DAT_140007164]...
...1115 LEA RCX,[DAT_140007148]
...111c LEA RAX,[DAT_140007164]
...1123 MOV qword ptr [RSP + local_18...]
...1128 CALL qword ptr [~>MSVCRT.DLL:~...]
...112e MOV dword ptr [DAT_14000714c]...
...1134 ADD RSP,0x38
...1138 RET

```



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The malware sample was successfully disassembled and analyzed using Ghidra, including its functions, strings, imports, and control flow.

The analysis helped identify potential malware behaviors and understand the program's functionality through assembly code and decompiled output.